Foreign Profit Repatriation and the Welfare Responses to the US–China Trade War: Evidence from Manufacturers in Vietnam

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Abstract

This paper studies how foreign ownership and profit repatriation shape Vietnam's welfare gains from the 2017–2019 US–China trade war. Using Vietnamese enterprise surveys covering all registered firms, I show that the production response came almost entirely from foreign-owned manufacturers: in sectors more exposed to US tariffs on Chinese goods, foreign-owned firms expanded employment and assets, while domestic firms contracted. I then estimate a quantitative trade model in which domestic and foreign-owned firms differ in productivity, fixed costs of sourcing and exporting, and the share of profits retained locally. A foreign demand shock matching the trade war's export response raises Vietnam's real expenditure by 7.95 percent, with two-thirds of the gain coming from higher real labor income. Profit repatriation matters surprisingly little: moving from full repatriation to full local retention raises the gain only from 7.94 to 8.10 percent, because retained foreign profits sustain less productive domestic firms and dampen real wage gains.

Keywords: US–China Trade War; Foreign Ownership; Profit Repatriation; Welfare Gains from Trade; Structural Estimation.

JEL Codes: F13; F12; F23; O53.

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1 Introduction

The US–China trade war starting in 2018 completely changed the landscape of international trade. The Trump administration imposed a series of tariffs on Chinese exports, raising the average import tariff from approximately 3 percent in early 2018 to over 20 percent by the end of 2019. During the same period, the average tariff level remained mostly constant for the rest of the world. As a result, global trade and value chains have been reallocating to some third countries producing close substitutes to Chinese exports, according to recent research (Alfaro and Chor, 2023; Fajgelbaum et al., 2024; Gopinath et al., 2025). Standard heterogeneous-firm theory predicts that such a shock reallocates resources toward the most productive firms and raises aggregate productivity (Melitz, 2003). Yet in many developing economies, the most productive and innovative firms are foreign-owned (Guadalupe et al., 2012; Xu et al., 2022), so much of the profit from this reallocation may accrue abroad rather than at home. Whether third countries gained from the trade war therefore depends not only on how much production they attracted, but also on who owns it. This paper asks three questions. Which firms in third countries responded to the US–China trade war? How large were the resulting welfare gains? To what extent would the gains change if foreign-owned firms repatriate their profits? I answer these questions for Vietnam, widely regarded as a leading beneficiary of the trade war, using a quantitative trade model with foreign ownership and profit repatriation.

I first provide evidence that the strong export and production response in Vietnam during the US–China trade war is almost entirely driven by foreign-owned manufacturers. The empirical analysis is conducted in two layers. At the product level, I use bilateral trade flows between Vietnam and the US/China/rest of the world (ROW) to show that (1) Vietnamese exports to the US and ROW increased significantly in final goods subject to the US import tariffs on Chinese exports and (2) Vietnamese imports from China increased significantly in intermediates and capital goods subject to the same set of tariffs. The results suggest that some manufacturers in Vietnam adjusted their production in response to the trade war by importing cheap intermediates from China, assembling them in Vietnam, and selling the final products to the US and ROW. Then, to examine the firm-level production response of Vietnam-based manufacturers during the trade war period, I utilize the Vietnam Enterprise Survey (VES) and construct a measure for sectoral exposure to the US–China trade war. Event study estimates show that foreign-owned manufacturers in exposed sectors raised their employment and assets at the onset of the tariff shock, while domestic firms contracted in employment and sales.

The empirical findings highlight the importance of complementarity between input sourcing and output exporting, as well as foreign ownership, in measuring the welfare gains of the trade war episode in Vietnam. Descriptively, foreign-owned and domestic manufacturers in Vietnam differ not only in their size distribution but also in their tendency to participate in trade activities as well as their effective income tax rates.[1] The trade participation and income tax gaps still hold when conditional on firm employment, suggesting that domestic and foreign-owned firms face different fixed costs of trade participation and retain profits locally to different extents. A quantitative trade model without foreign ownership would miss these margins and risk miscalculating the gains from foreign demand shocks.

To correctly quantify Vietnam's welfare gains from the US–China trade war, I develop and estimate a quantitative model of trade participation with foreign ownership and profit repatriation based on the global sourcing model of Antràs et al. (2017). There are several novel features of my model relative to theirs. First, I introduce the margin of exporting into the model. Final goods producers choose whether to source their inputs and export their products to the world market given their core productivity and fixed costs of participating in trade. The resulting profit function exhibits complementarity between the sourcing and exporting decisions, which echoes the empirical finding from the product-level analysis. Second, I introduce the distinction between domestic and foreign ownership. Final goods producers are either domestic or foreign-owned and draw their productivity and fixed costs from separate distributions, consistent with the findings from the firm-level analysis. Additionally, only a proportion of profits from foreign-owned firms are retained in the host country, which is calibrated to match Vietnam's GNI-to-GDP ratio in 2017. All the remaining model parameters are then estimated via the simulated method of moments. The estimates confirm substantial differences in fixed costs by firm ownership, where the fixed costs of sourcing and exporting for foreign-owned manufacturers are only 11 percent and 21 percent of those for domestic manufacturers.

With the baseline parameters, I conduct a counterfactual exercise to elucidate the welfare gains from the US–China trade war. Specifically, I impose a positive foreign demand shock on the estimated model, which induces an average export response of the same magnitude as the one by the US trade war tariffs estimated from the data. I show that the shock induces an 7.95% increase in Vietnam's real expenditure, with 66 percent of the change from real wage gains and the rest from the retained profits of firms.

[1]Many MNEs have been subject to tax disputes with the Vietnamese government in recent years, e.g. Adidas, Suntory, Unilever, etc. A pronounced case is Coca-Cola, which has been operating in Vietnam since 1993 but only started reporting positive profits from 2013 onward. In 2019, the General Department of Taxation required the company to pay over 35 million USD in fines and tax arrears (Lam, 2021).

The welfare gains from the tariff shock improve only moderately when the extent of foreign profit repatriation is reduced. I found that the welfare gains would move from 7.94% under full repatriation to 8.10% under full profit retention. The surprisingly small difference is due to the adjustment in real wages: despite the inclusion of foreign profits in total household income when repatriation is prohibited, the real wage gains become smaller when higher income sustains more unproductive domestic firms in equilibrium, which offsets part of the welfare gains from retained foreign profits. A decomposition exercise also shows that the differences in fixed costs between domestic and foreign-owned firms can explain the majority of the trade participation gap between the two groups of firms.

This paper makes contributions to at least three strands of literature. The first is about the impact of the 2017–2019 US–China trade war on third countries. Cross-country evidence (Alfaro and Chor, 2023; Fajgelbaum et al., 2024) suggests that Trump tariffs led to the restructuring of global value chains, with notable countries like Vietnam and Mexico gaining US import market shares. Recent empirical studies focusing on these countries show that the trade war episode led to job creation, improved labor conditions, and reallocation of workers into the formal manufacturing sector (Malesky and Mosley, 2021; Nguyen and Lim, 2023; Utar et al., 2025; Chen et al., 2025; Rotunno et al., 2026; Mayr-Dorn et al., 2026). On the other hand, some studies examine Chinese-linked supply chains in Vietnam and find most parts are legitimate input usage rather than illegal rerouting (Freund et al., 2023; Iyoha et al., 2026).[2] To the best of my knowledge, this paper is the first to provide a quantitative general equilibrium framework to study the welfare implications of the US–China trade war in an emerging economy.

This paper also contributes to the extensive literature on trade liberalization and reallocation in developing countries (Atkin and Khandelwal, 2020; Atkin and Donaldson, 2022). In particular, McCaig and Pavcnik (2018) and McCaig et al. (2025) study the impact of the 2001 US–Vietnam bilateral trade agreement (BTA) on the Vietnamese economy and show that it expanded the formal sector and encouraged foreign firm entry by improving exporting opportunities. My research echoes their empirical findings about the reallocation of resources toward foreign-owned firms in Vietnam and further digs into the welfare gains from a foreign demand shock in the presence of foreign profit repatriation in a quantitative framework.

[2] Iyoha et al. (2026) use Vietnam's bill-of-lading data to measure illegal trade rerouting, in which Chinese products are relabeled as Vietnamese, and legitimate Chinese intermediate content embedded in Vietnam's exports. Together, these channels account for 27.1% of the surge in Vietnam's exports to the US over 2018–2022, with potential rerouting contributing 10.5 percentage points and Chinese imported content 16.6 points.

Last, this paper also relates to the quantitative assessment of international corporate tax and profit shifting (Wang, 2020; Ferrari et al., 2022; Dyrda et al., 2024). Rather than studying the welfare implications of different tax regimes, I focus on studying how the welfare gains from positive tariff shocks would differ under different levels of foreign profit repatriation.

The rest of the paper is organized as follows. Section 2 presents the data sources and key descriptive facts of Vietnam-based manufacturers. Section 3 studies Vietnam's trade and production response to the US–China trade war episode. Based on the empirical findings, Section 4 presents the theoretical framework. Section 5 describes the details of the model parameter estimation. With the estimated model, Section 6 conducts the counterfactual exercise and welfare analysis. Section 7 concludes.

2 Data and Descriptive Facts

In this section, I introduce the main datasets and provide three key descriptive facts about foreign-owned manufacturers in Vietnam. Some of these facts have been documented before by other papers, e.g., McCaig et al. (2025). Nonetheless, the objective is to highlight the distinction between domestic and foreign-owned firms in Vietnam and guide the intuition for the reduced-form analysis and theoretical framework in the following sections.

2.1 Data

For the majority of my empirical analysis, I use the Vietnam Enterprise Survey (VES) from 2015 to 2021. It includes all officially registered firms in Vietnam and provides data on firm ID, location, ISIC 4-digit industry classification, and balance sheet information, offering a comprehensive view of the corporate landscape. Importantly, the survey asks about the firms' main capital source country. I complement the VES with Vietnam Technology and Competitiveness Survey (TCS) data from 2009 to 2014, which covers a subset of firms from the VES and provides more trade-related information such as details on firms' sourcing and exporting activities. Additionally, I employ the CEPII BACI Database (Gaulier and Zignago, 2010) for bilateral trade flows by HS 6-digit product code.

2.2 Descriptive Facts on Vietnamese Manufacturers

Fact 1. Foreign-owned manufacturers in Vietnam are larger than domestic ones. In Table 1, I first examine the main outcomes for active manufacturers in Vietnam by ownership category. Foreign-owned manufacturers are defined as manufacturers in the VES data reporting 100 percent of their capital to be sourced from abroad, and domestic manufacturers are the non-foreign-owned manufacturers in the VES data.[3] On average, foreign-owned manufacturers in Vietnam are about ten times larger than domestic manufacturers in employment, assets, and sales. In Panel (a) of Figure 1, I plot the employment distribution in 2017. Domestic manufacturers are mostly small in size, while foreign-owned manufacturers concentrate more in the top percentiles of the distributions.

Table 1: Summary Statistics of Vietnamese Manufacturers in 2017

	Employment	Assets	Sales	Exporter share	Importer share
Domestic					
Mean	44	1440	1577	0.049	0.047
Median	7	122	76	0.000	0.000
SD	287	14357	20606	0.216	0.211
Observation	64664	64664	64664	64664	64664
Foreign-owned					
Mean	542	12680	18272	0.633	0.644
Median	106	2001	1865	1.000	1.000
SD	2132	139045	287980	0.482	0.479
Observation	6352	6352	6352	6352	6352

Note: The sample comprises Vietnamese manufacturers with positive employment, sales, and assets in the 2017 Vietnam Enterprise Survey. The unit of assets and sales is 1000 USD.

Fact 2. Foreign-owned manufacturers participate more in trade activities than domestic manufacturers conditional on firm size.
In Table 1, only approximately five percent of domestic manufacturers engage in import and export on average, while the numbers are over 60 percent for foreign-owned manufacturers. Furthermore, I plot the shares of domestic and foreign-owned manufacturers in each percentile in Panel (b) of Figure 1. Foreign-owned manufacturers are much more likely to participate in both import and export activities across all employment percentiles.

Fact 3. Effective tax rates are lower for foreign-owned manufacturers conditional

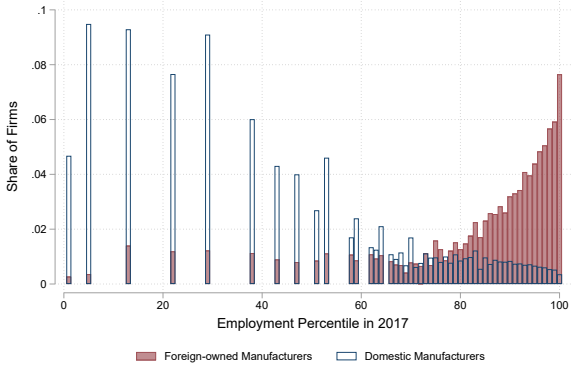
[3]Most foreign direct investments in Vietnam are greenfield investments. In Figure A1 of Appendix A, I show that foreign joint ventures in Vietnam have a very small presence in terms of both firm count and employment.

on firm size.

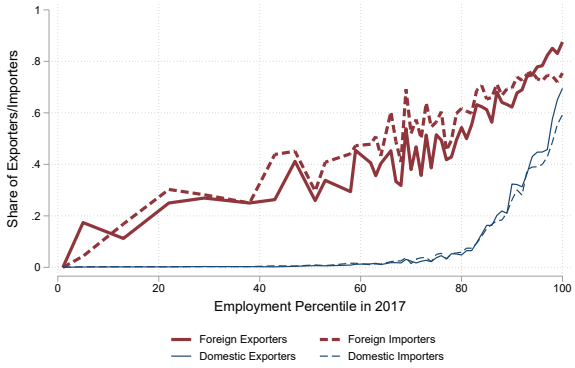
Panel (c) of Figure 1 shows the median values of effective tax rates[4] across employment percentiles for both types of firms. Throughout the size distribution, median domestic firms' effective tax rates are 20%, the same level as Vietnam's official income tax rate. On the other hand, the median tax rates for foreign-owned firms are lower than 20% and decrease with firm size.

Figure 1: Employment, Trade Participation, and Effective Tax Rates in 2017

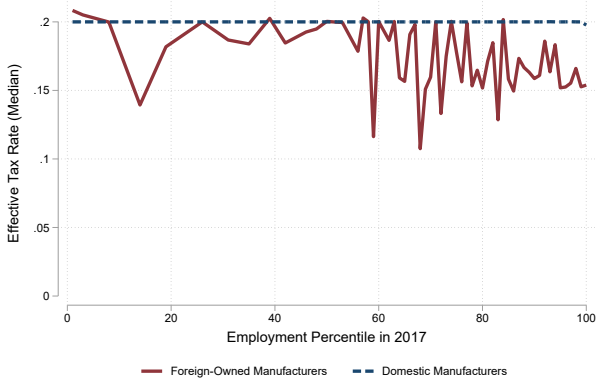
(a) Employment Distribution



(b) Trade Participation



(c) Effective Tax Rate



Note: The sample comprises Vietnamese manufacturers with positive employment, sales, and assets in the 2017 Vietnam Enterprise Survey. Employment percentiles pool foreign-owned and domestic firms. Panel (a) reports firm counts in each percentile divided by total firm counts within each ownership group. Panel (b) reports the fractions of firms with positive exports or imports within each percentile and ownership group. Panel (c) reports the median ratio of income tax expenses to earnings before tax within each percentile and ownership group, restricting the sample to firms with positive earnings and tax expenses.

[4]The effective tax rates are defined as the income tax expenses divided by total earnings before tax. Manufacturers with non-positive tax expenses or non-positive earnings are not included in the calculation.

3 Reduced-Form Analysis

In this section, I exploit the US–China trade war in 2018–2019 as a positive foreign demand shock to Vietnam and study the overall trade response at the product level and the manufacturing production response at the sector level in Vietnam. The product-level results highlight that the Trump tariffs on Chinese exports triggered a rise in Vietnamese exports and imports. Moreover, the export response is stronger in final goods to the US, and the import response is stronger in intermediates from China. At the sector level, I also see that the exposed sectors in Vietnam increased their employment and assets, predominantly driven by foreign-owned manufacturers.

3.1 Trade Response

Using the bilateral trade flow data from CEPII's BACI Database, I examine how the growth rate of Vietnamese trade with all countries/US/China/rest of the world (ROW) responded to the US–China trade war at the product level. The following regression specification is adopted:

$$\begin{aligned}\Delta_{17,19} \log Y_p^{k,VN} =& \beta_0 + \beta_1 \, \Delta_{17,19} \log(1 + \tau_p^{US,CN}) + \beta_2 \, \Delta_{17,19} \log(1 + \tau_p^{CN,US}) \\ &+ \beta_3 \, \Delta_{17,19} \log(1 + \tau_p^{US,VN}) + \beta_3 \, \Delta_{13,15} \log Y_p^{k,VN} + \epsilon_p, \end{aligned} \tag{1}$$

where $\Delta_{17,19} \log Y_p^{k,VN}$ is the difference in log trade values of an HS 6-digit product p between Vietnam and country k from 2017 to 2019, $\Delta_{17,19} \log(1+\tau_p^{US,CN})$ is the change in the log US import tariff on exports of product p from China, $\Delta_{17,19} \log(1+\tau_p^{CN,US})$ is the change in the log Chinese import tariff on exports of product p from the US, and $\Delta_{17,19} \log(1 + \tau_p^{US,VN})$ is the change in the log US import tariff on exports of product p from Vietnam. One caveat is that products that have zero values in 2017 or 2019 won't be included under this specification.[5]

The changes of log tariffs in 2017-2019 are summarized in Table 2. On average, US import tariffs on Chinese exports increased by 15 percentage points in this period, while they increased merely by 0.2 percentage points for Vietnamese exports. In response, the Chinese retaliatory tariffs on imports from the US increased by 11 percentage points. The standard deviations of the log change of US–China, China–US, and US–Vietnam import

[5]In Appendix B.1, I run a robustness check using the Davis–Haltiwanger growth rate measure to incorporate the extensive margin, which produces similar results.

tariffs are about 0.10, 0.09, and 0.02, respectively.

Table 2: Summary of Trade War Tariffs

	$\Delta_{17,19}\log(1+\tau_p^{US,CN})$	$\Delta_{17,19}\log(1+\tau_p^{CN,US})$	$\Delta_{17,19}\log(1+\tau_p^{US,VN})$
Mean	0.154	0.113	0.002
Median	0.214	0.118	0.000
SD	0.096	0.088	0.020

Note: Sample is all HS 6-digit products. The US–China trade war tariffs in 2017-2019 are obtained from Fajgelbaum et al. (2020).

The corresponding estimates of β_1 for Vietnamese import and export values by destination in Equation (1) are shown in Table 3. On the one hand, Vietnamese exports for a given product responded significantly to the rise in US–China import tariffs. On the other hand, Vietnamese imports also rose in response to the US–China import tariffs, particularly imports from China. In Column (1) of Panel (a), the average increase in US tariffs on Chinese exports (15%) led to the overall Vietnamese export growth by 0.14 log points (15%).[6] In Columns (2)-(4) of Panel (a), the same tariff change increased the growth of exports to the US, China, and ROW by 25%,[7] 15 % (insignificant),[8] and 14 %,[9] respectively. On the other hand, the same tariff increase led to average Vietnamese import growth of 5% overall (insignificant),[10] 7% from the US (insignificant),[11] 11% from China,[12] and 2% from ROW (insignificant)[13] in Column (1)-(4) of Panel (b). Using the classification by Broad Economic Categories (BEC) from the United Nations Statistics Division, I further separate the products into intermediates and final goods and run the same specification in Tables B1 and B2 of Appendix B.1. It can be seen that the response of exports to the US is driven mostly by final goods, while the response of imports from China is driven mainly by the surge in intermediate and capital goods.

[6] $0.949 \times 0.15 \approx 0.14; e^{0.14} - 1 \approx 0.15$.
[7] $1.479 \times 0.15 \approx 0.22; e^{0.22} - 1 \approx 0.25$.
[8] $0.932 \times 0.15 \approx 0.14; e^{0.14} - 1 \approx 0.15$.
[9] $0.851 \times 0.15 \approx 0.13; e^{0.13} - 1 \approx 0.14$.
[10] $0.335 \times 0.15 \approx 0.05; e^{0.05} - 1 \approx 0.05$.
[11] $0.496 \times 0.15 \approx 0.07; e^{0.07} - 1 \approx 0.07$.
[12] $0.711 \times 0.15 \approx 0.11; e^{0.11} - 1 \approx 0.11$.
[13] $0.147 \times 0.15 \approx 0.02; e^{0.02} - 1 \approx 0.02$.

Table 3: Effect of Trade War Tariffs on Vietnamese Trade Value

(a) Export Value

	(1) Total Export	(2) Export to US	(3) Export to China	(4) Export to ROW
$\Delta \log(1+\tau_p^{US,CN})$	0.949	1.479	0.932	0.851
	(0.311)	(0.531)	(0.725)	(0.314)
Observations	4217	1785	1664	4155
R^2	0.006	0.020	0.005	0.003

(b) Import Value

	(1) Total Import	(2) Import from US	(3) Import from China	(4) Import from ROW
$\Delta \log(1+\tau_p^{US,CN})$	0.335	0.496	0.711	0.147
	(0.236)	(0.415)	(0.288)	(0.252)
Observations	4871	2669	3976	4785
R^2	0.001	0.001	0.002	0.003

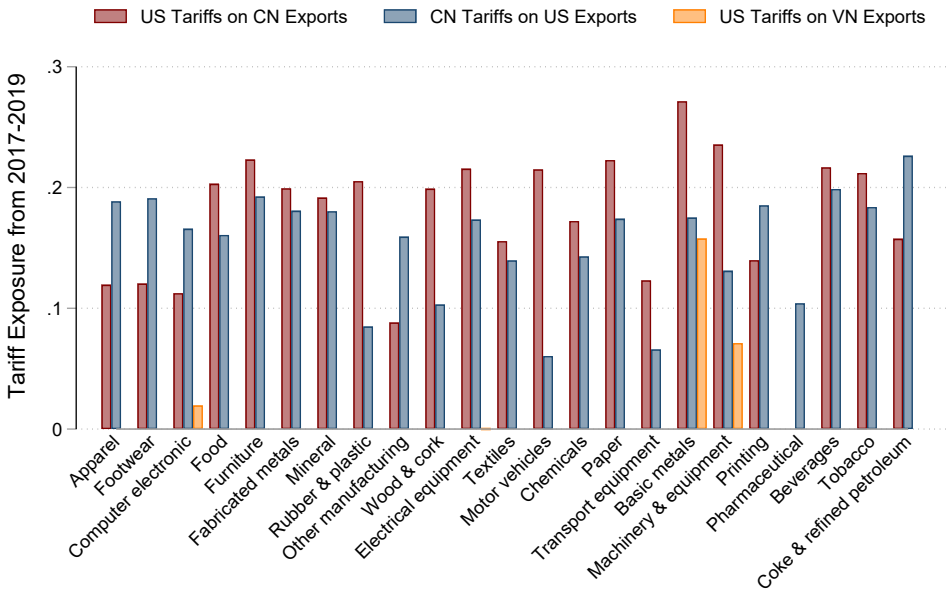
Note: Standard errors are clustered at the HS 6-digit product level. ROW stands for the rest of the world.

3.2 Production Response

Because the VES data do not record the products that manufacturers produce, I have to define a measure that aggregates the product-level tariffs to the sector level. For each 4-digit ISIC manufacturing sector in Vietnam, I define its exposure to US import tariffs as a simple average of the product-level tariffs, with the weights given by the value shares of Vietnamese exports to the US for every product within the given sector.[14] On average, manufacturers in our firm sample in 2017–2019 were exposed to 14, 11, 0.04 percent increases in US import tariffs on Chinese exports, Chinese import tariffs on US exports, and US import tariffs on Vietnamese exports, respectively (see Table B4 of Appendix B.2). The average tariff exposure by 2-digit manufacturing sector is illustrated in Figure 2. There are rich variations in the exposure to the US tariffs on Chinese exports and the Chinese retaliatory tariffs, while the exposure to tariffs on Vietnamese exports concentrates mostly in the basic metals sector due to the steel and aluminum tariffs imposed by the Trump administration.

[14] Details of the measure are provided in Appendix B.2.

Figure 2: Tariff Exposure by Manufacturing Industry in Vietnam



Note: The figure illustrates the tariff exposure measure by ISIC 2-digit manufacturing industry. Industries are ordered by their total employment (from left to right) in Vietnam in 2017.

With the exposure measure, I examine Vietnamese manufacturers' production response by firm ownership using a triple difference-in-differences (DIDID) specification:

$$\begin{aligned}\log Y_{ij(4)kt} =& \beta_0 \, \Delta_{17,19} \log(1+\tau^{US,CN}_{j(4)}) \times Post_t + \beta_1 \, \Delta_{17,19} \log(1+\tau^{US,CN}_{j(4)}) \times Post_t \times Foreign_i \\ &+ X_{ij(4)t} + \alpha_i + \alpha_t + \alpha_{j(2)t} + \alpha_{kt} + \epsilon_{ij(4)kt}, \end{aligned} \tag{2}$$

where i indexes a manufacturer, $j(4)$ indexes an ISIC 4-digit industry, k indexes ownership type, and t indexes a year in 2015–2021. The outcomes of interest, Y, include firm employment, total assets, and total sales.[15] $Foreign_i$ indicates whether i is foreign-owned. $Post_t$ indicates years after 2017. $X_{ij(4)t}$ is a set of control variables involving the analogous interaction terms of Chinese retaliation tariff change and US direct tariff change with foreign ownership and post. $\{\alpha\}_{ij(2)kt}$ are firm, year, ISIC 2-digit industry by year, and foreign ownership by year fixed effects. Coefficient β_0 captures domestic producers' average production response to Trump tariffs, and β_1 shows the additional response for foreign-owned firms.[16]

[15] To avoid overrepresentation of extreme observations, Monetary variables are winsorized at their annual 1st and 99th percentiles.

[16] The coefficients for $\Delta_{17,19} \log(1+\tau^{US,CN}_{j(4)})$ and $Post_t$ are absorbed by the fixed effects α_i and α_t.

In Table 4, the estimates of β_0 and β_1 are reported. Domestic and foreign-owned manufacturers exhibit a stark contrast in their response to the tariff shock: an average change in tariff exposure (14%) leads to a 3% drop in employment and a 16% drop in total sales for domestic producers, while their foreign counterparts gain 8% and 5% in employment and assets, respectively.[17] The results are similar when restricting to incumbent firms and using Vietnam's export shares as tariff weights in Appendix B.2.

Table 4: Effect of Trade War Tariffs on Employment and Sales

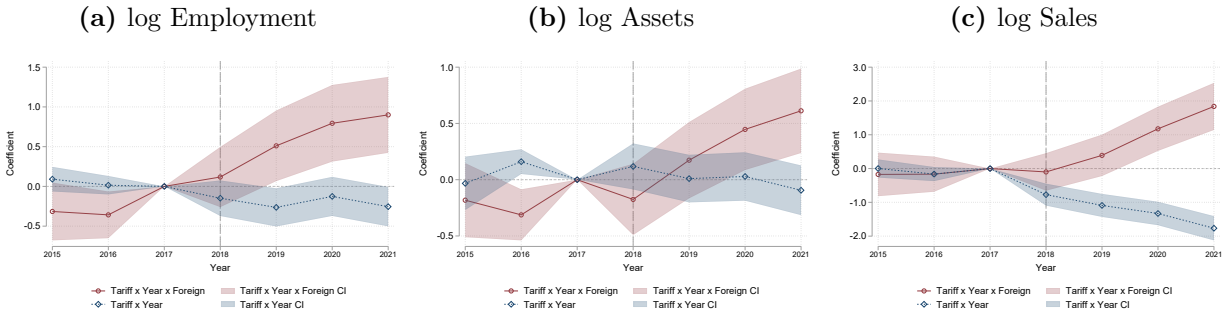
	(1) log Employment	(2) log Assets	(3) log Sales
$\Delta \log(1+\tau_{j(4)}^{US,CN}) \times Post_t$	-0.227	-0.025	-1.168
	(0.112)	(0.107)	(0.162)
$\Delta \log(1+\tau_{j(4)}^{US,CN}) \times Post_t \times Foreign_i$	0.780	0.411	0.908
	(0.196)	(0.162)	(0.278)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes
Foreign × Year FE	Yes	Yes	Yes
Total tariff effect: Foreign	0.554	0.386	-0.260
	(0.200)	(0.163)	(0.279)
Observations	457414	457414	457414
R-squared	0.894	0.921	0.882

Note: The sample consists of Vietnamese manufacturers present in the Vietnam Enterprise Survey from 2015 to 2021 with positive employment, assets, and sales. Monetary variables are winsorized at the 1st and 99th percentiles. Regressions include firm, year, two-digit industry-by-year, and foreign-status-by-year fixed effects, and control for changes in Chinese tariffs on U.S. goods and U.S. tariffs on Vietnamese goods, interacted with foreign status and a post-2017 indicator. Standard errors are clustered at the firm level.

To illustrate the dynamic effects, Figure 3 shows the analogous estimates of tariff exposures with year dummies. After the Trump tariff shock, foreign-owned firms continue to grow in all three outcomes, while domestic firms stay stagnant or contract. The DIDID estimates are marginally significant for employment and assets pre-2017, but their combinations with the tariff-year estimates (overall effects of tariff exposures on foreign-owned firms) are not.

[17] The linear combination of β_0 and β_1 for log sales is not statistically different from zero.

Figure 3: Event Study



Note: The figures report event-study estimates for Vietnamese manufacturing firms in the VES panel over 2015–2021. Navy lines show coefficients on tariff exposure for domestic firms, and maroon lines show the additional coefficients for foreign firms. All coefficients are normalized to zero in 2017. Shaded bands represent 95% confidence intervals.

3.3 Discussion of Empirical Findings

Overall, the reduced-form analysis provides a complete picture of how Vietnam responded to the trade war in 2017–2021. The product-level analysis indicates that Vietnam increased its imports from China, in particular in capital and intermediate goods, and its exports to the US and the rest of the world, in particular in final goods. This is supporting evidence of Vietnam-based firms reacting to the shock by sourcing cheaper inputs, assembling them locally, and exporting the final products to the world market. Furthermore, the firm-level analysis focusing on manufacturers shows that the positive increases in employment and assets were mainly coming from the foreign-owned manufacturers. The significant heterogeneity in production response highlights the importance of taking foreign ownership seriously. These findings are consistent with Iyoha et al. (2026), which shows that Chinese imported content, rather than illegal rerouting activities, plays an important role in Vietnam's export surge to the US in 2018–2022.

One important caveat of the regression analysis is that it captures only the relative, not the absolute, effects of tariff exposure. A foreign demand shock at the scale of the trade war has a general equilibrium effect on prices and wages, thus affecting products or sectors not directly exposed to the episode. To understand the overall welfare effect of a foreign demand shock, I develop a theoretical framework of firm participation with foreign ownership and profit repatriation in the next section.

4 Model of Trade Participation With Firm Ownership

The reduced-form analysis in the previous section confirms that firms' response to foreign demand shocks can differ starkly with their ownership in a developing country. To further understand the welfare implications of these responses, I develop a static framework of trade participation with foreign ownership and profit repatriation based on the global sourcing model of Antràs et al. (2017). In the following, I first describe the overall environment and then introduce the consumer demand and goods supply in detail and lay out the general equilibrium conditions of the model.

4.1 Environment

There are two countries, home and abroad,[18] each with a representative consumer and a fixed mass of final good manufacturers. The consumer offers labor inelastically and consumes manufacturing varieties. Each final good manufacturer produces a single variety and engages in monopolistic competition. To produce a variety, the firm employs domestic workers to assemble a unit measure of inputs, which can be sourced either from home or from abroad. After the products are produced, they can be sold either locally or overseas. Given the local and foreign demand, firms' variable profits are determined by their own core productivity, which is already known before they make any production decisions.

On top of the standard setting, each firm has an ownership status: domestic or foreign. The ownership status matters in three key aspects. First, firms have to pay ownership-specific operation costs to become active in any production activities. They prefer to stay inactive if they cannot make positive profits net of the costs given their own productivity and consumer demand. The ownership-specific operation costs thus lead to self-selection into production by ownership and different productivity distributions for active domestic and foreign-owned firms. Second, a firm has to pay ownership-specific fixed costs of sourcing and exporting. This generates self-selection into import and export activities. Last, domestic firms are owned by the representative household, and hence their net profits contribute wholly to domestic consumption; however, I assume that foreign-owned firms contribute only a constant share of their net profits to consumption in the home country. Later in Section 6, I vary the contributed share and examine the welfare implications for the home country.

Before proceeding to the model details, I summarize the timing of firm decisions. First,

[18] I do not call the other country "foreign" to prevent confusion with "foreign-owned" firms.

all final good manufacturers draw their core productivity. Then, they decide whether to pay the fixed operation costs and start producing their product. Once they become active in production, they choose whether to participate in trade activities, including sourcing some of their inputs and exporting their products abroad. Finally, firm profits are realized given their core productivity and trade participation decisions.

4.2 Consumer Demand

The representative consumer in country $k \in \{h, a\}$ has constant-elasticity-of-substitution (CES) utility over differentiated manufacturing final goods:[19]

$$U_k = \left(\int_{\Omega} q_k(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \tag{3}$$

where σ is the elasticity of substitution over varieties.

The representative consumer in each country maximizes her utility by choosing how much to consume for each variety given her budget constraint. This leads to the consumer demand for each variety ω:

$$q_k(\omega) = E_k P_k^{\sigma-1} p_k(\omega)^{-\sigma}, \tag{4}$$

where E_k is the total manufacturing expenditure and P_k is the aggregate manufacturing price index, measured in country k's own currency. Following Cosar et al. (2016), I assume that the home country is a small open economy and imports a fixed variety of final goods from abroad, N_a. As a result, the home-currency price index for imported final goods is $\tau^M \epsilon \left[\int_0^{N_a} p_a(\omega)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}$, where ϵ is the exchange rate (home-currency units per currency unit abroad) and $p_a(\omega)$ is the free-on-board (FOB) price of imported variety $\omega \in [0, N_a]$. Analogously, the home-currency price index for local final goods is $\left[\int_{\Omega_h} p_h(\omega)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}$, where Ω_h is endogenously determined by the measure of active manufacturers at home. Since the measure of foreign varieties and their FOB prices are exogenous to the model, $\left[\int_0^{N_a} p_a(\omega)^{1-\sigma} d\omega\right]^{\frac{1}{1-\sigma}}$ is normalized to one by the choice of foreign currency for simplicity.

[19] The main focus of this model is the home country, i.e., Vietnam. To avoid cumbersome notation, the country subscript below is mostly omitted and used only when necessary. The setup for the other country in the model is analogous.

Therefore, the aggregate price index at home is

$$P_h = \left[\int_{\Omega_h} p_h(\omega)^{1-\sigma} d\omega + (\tau^M \epsilon)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \tag{5}$$

Following the convention in Antràs et al. (2017), I define a residual demand term here, which will be useful later:

$$B_k \equiv c_\sigma E_k P_k^{\sigma-1}, \qquad c_\sigma \equiv \frac{1}{\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{1-\sigma}. \tag{6}$$

4.3 Goods Supply

Each final good manufacturer at home makes a unique variety by hiring workers to assemble a continuum of intermediate inputs. The firms are indexed by their ownership status $i \in \{d, f\}$, i.e., domestic or foreign. The marginal cost of a manufacturer of ownership status i with core productivity φ is thus given by:

$$c_i(\varphi) = \frac{1}{\varphi} w_h^\gamma \left(\int_0^1 z_i(\nu, \varphi)^{(1-\rho)} d\nu \right)^{\frac{1-\gamma}{1-\rho}}, \tag{7}$$

where γ is the labor expenditure share and ρ is the substitution elasticity between inputs. w_h is the wage rate of manufacturing workers at home, and $z_i(\nu, \varphi)$ is the price of a given input ν for a firm of ownership i and productivity φ.

Input producers are assumed to use only labor and engage in perfect competition. Therefore, the price of input ν that a manufacturer pays is:

$$z_i(\nu, \varphi) = \begin{cases} e_h(\nu) w_h & \text{if } J_i^M(\varphi) = 0 \\ \min\{e_h(\nu) w_h, e_a(\nu) \epsilon \tau^M w_a\} & \text{if } J_i^M(\varphi) = 1, \end{cases} \tag{8}$$

where τ^M is the iceberg trade cost of importing a unit of input ν from abroad, $\{e_h, e_a\}$ are the unit labor requirements of the input at home and abroad, and $J_i^M(\varphi)$ is an indicator for whether a manufacturer with ownership i and productivity φ engages in import activities. Following Eaton and Kortum (2002), the unit labor requirements for each country are assumed to be drawn from the Fréchet distribution:

$$\Pr(e_k(\nu) \geq e) = \exp(-T_k e^\theta). \tag{9}$$

Then, given its import status J_i^M, the import share of the producer is:

$$\chi_i^M(\varphi) = \begin{cases} 0 & \text{if } J_i^M(\varphi) = 0 \\ \frac{\epsilon^{-\theta} T_a (\tau^M w_a)^{-\theta}}{T_h w_h^{-\theta} + \epsilon^{-\theta} T_a (\tau^M w_a)^{-\theta}} & \text{if } J_i^M(\varphi) = 1. \end{cases} \tag{10}$$

Taking the consumer demand at home and abroad as given, the final good manufacturer with ownership i and productivity φ engages in monopolistic competition with other firms. This leads to the constant markup pricing rule, i.e., $p_i(\varphi) = \frac{\sigma}{\sigma-1} c_i(\varphi)$. It is easy to show that the variable and net profit of the firm can be derived as follows:

$$\begin{aligned} \pi_v(\varphi, J_i^M, J_i^X) &= \left(\frac{\varphi}{w_h^\gamma \beta}\right)^{\sigma-1} \left(\underbrace{\frac{T_h}{w_h^\theta}}_{\Theta_h^M} + J_i^M \epsilon^{-\theta} \underbrace{\frac{T_a}{(\tau^M w_a)^\theta}}_{\Theta_a^M} \right)^{\frac{(1-\gamma)(\sigma-1)}{\theta}} \left(\underbrace{B_h}_{\Theta_h^X} + J_i^X \epsilon^{\sigma} \underbrace{\frac{B_a}{(\tau^X)^{\sigma-1}}}_{\Theta_a^X} \right), \\ \pi_i(\varphi, J_i^M, J_i^X) &= \pi_v(\varphi, J_i^M, J_i^X) - w_h f_i^O - J_i^M w_h f_i^M - J_i^X w_h f_i^X \quad \forall i \in \{d, f\}, \end{aligned} \tag{11}$$

where β is some constant,[20] $J_i^X(\varphi)$ is an indicator for whether a manufacturer with ownership i and productivity φ engages in export activities, $\{\Theta_h^M, \Theta_a^M\}$ are defined as the **sourcing potential** at home and abroad, and $\{\Theta_h^X, \Theta_a^X\}$ are defined as the **market potential**. Furthermore, manufacturers have to pay ownership-specific operation costs f_i^O to stay active in production. They also have to pay sourcing and exporting fixed costs $\{f_i^M, f_i^X\}$ if they decide to participate in either of these activities; these costs also differ by ownership status.

The following proposition shows that the trade participation decisions are complementary to each other and to firms' own productivity. The proof is provided in Appendix C.

Proposition 1 *π_i has increasing differences in (φ, J_i^M), (φ, J_i^X), and (J_i^M, J_i^X). Therefore,*

1. *more productive manufacturers are more likely to participate in import and export, and*
2. *improvement in market potential abroad weakly increases manufacturers' export and import participation.*

[20] $\beta \equiv \frac{1}{\gamma^\gamma (1-\gamma)^{1-\gamma}} \Gamma\left(\frac{\theta-\rho+1}{\theta}\right)^{\frac{1-\gamma}{1-\rho}}$.

Last, the export share of firm (i, φ) can be derived as the following:

$$\chi_i^X(\varphi) = \begin{cases} 0 & \text{if } J_i^X(\varphi) = 0 \\ \frac{\epsilon^\sigma B_a (\tau^X)^{1-\sigma}}{B_h + \epsilon^\sigma B_a (\tau^X)^{1-\sigma}} & \text{if } J_i^X(\varphi) = 1. \end{cases} \tag{12}$$

4.4 Equilibrium

Fixed numbers of potential entrants $\{N_d, N_f\}$ are present at home with knowledge of their own core productivity φ. A manufacturer will pay the associated operation costs f_i^O to stay active in production if its realized profit in Equation (11) is weakly positive. Therefore, the domestic and foreign-owned manufacturers who earn zero profits will pin down the respective equilibrium cutoff productivities $\{\underline{\varphi}_d^*, \underline{\varphi}_f^*\}$:[21]

$$\pi_i(\underline{\varphi}_i^*, J_i^{M*}, J_i^{X*}) = 0 \quad \forall i \in \{\text{d}, \text{f}\}. \tag{13}$$

The trade balance condition requires that the equilibrium exchange rate ϵ^* adjusts to equalize the total amount of exports from home and the total amount of imports from abroad plus repatriated profits:

$$\begin{aligned} \sum_{i \in \{d,f\}} N_i \int_{\underline{\varphi}_i}^{\infty} \underbrace{\chi_i^X(\varphi) \sigma \pi_v(\varphi, J_i^{M*}, J_i^{X*})}_{\text{Exported final goods}} \, dG(\varphi) = \underbrace{\frac{B_h^*}{c_\sigma} (\tau^M \epsilon^*)^{1-\sigma}}_{\text{Imported final goods}} \\ + \sum_{i \in \{d,f\}} N_i \int_{\underline{\varphi}_i^*}^{\infty} \underbrace{\chi_i^M(\varphi)(1-\gamma)\,(\sigma-1)\pi_v(\varphi, J_i^{M*}, J_i^{X*})}_{\text{Imported inputs}} \, dG(\varphi) \\ + \sum_{i \in \{d,f\}} N_i \int_{\underline{\varphi}_i^*}^{\infty} \underbrace{(1-\lambda_i)\,\pi_i(\varphi, J_i^{M*}, J_i^{X*})}_{\text{Repatriated profits}} \, dG(\varphi), \end{aligned} \tag{14}$$

where $G(\varphi)$ is the unconditional distribution of firm productivities, and λ_i is a number in $[0, 1]$, representing the share of profits that firms of ownership i contribute to expenditure in the home country. For the baseline specification, λ_d is set to one, and λ_f is calibrated to match Vietnam's pre-trade-war GNI-to-GDP ratio. In Section 6, I consider the welfare implication when λ_f varies.

Because the trade balance condition holds, aggregate expenditure at home equals ag-

[21] In the next section, sourcing and exporting fixed costs are parameterized as draws from log-normal distributions, so the cutoffs alone do not determine entry completely. This condition here serves as a theoretical approximation.

gregate income at home, given by the sum of labor income and retained net profits from firms:

$$E_h^* = \underbrace{w_h L_h}_{\text{Labor income}} + \sum_{i \in \{d,f\}} N_i \int_{\underline{\varphi}_i^*}^{\infty} \underbrace{\lambda_i \, \pi_i(\varphi, J_i^{M*}, J_i^{X*})}_{\text{Retained profits}} \, dG(\varphi). \tag{15}$$

The goods market clearing condition pins down the equilibrium aggregate price index P_h^* and thus the equilibrium residual demand at home B_h^* following Equation (6).

$$E_h^* = \underbrace{\frac{B_h^*}{c_\sigma}(\tau^M \epsilon^*)^{1-\sigma}}_{\text{Imported final goods}} + \sum_{i \in \{d,f\}} N_i \int_{\underline{\varphi}_i^*}^{\infty} \underbrace{(1 - \chi_i^X(\varphi))\sigma \pi_v(\varphi, J_i^{M*}, J_i^{X*})}_{\text{Local final goods}} \, dG(\varphi). \tag{16}$$

Workers supply their labor inelastically given wages. The following labor market clearing must hold in equilibrium:

$$\begin{aligned} w_h L_h = \sum_{i \in \{d,f\}} N_i \int_{\underline{\varphi}_i^*}^{\infty} & \underbrace{w_h (f_i^O + J_i^{M*} f_i^M + J_i^{X*} f_i^X)}_{\text{Wages paid for fixed costs}} \\ + & \underbrace{\gamma(\sigma - 1)\pi_v(\varphi, J_i^{M*}, J_i^{X*})}_{\text{Wages paid for final goods production}} + \underbrace{(1 - \chi_i^M(\varphi))(1 - \gamma)\,(\sigma - 1)\pi_v(\varphi, J_i^{M*}, J_i^{X*})}_{\text{Wages paid for local input production}} \, dG(\varphi). \end{aligned} \tag{17}$$

By Walras' law, one can show that Equations (14), (15), and (16) together imply Equation (17). Hence, the wage is normalized to unity, and labor is used as the numeraire.

Last, the set of parameters in the model is defined as Λ and includes the demand elasticity σ, the input trade elasticity θ, the labor expenditure share γ, the shares of retained profits $\{\lambda_d, \lambda_f\}$, the sourcing and market potential abroad $\{\Theta_a^M, \Theta_a^X\}$, the fixed operation costs $\{f_d^O, f_f^O\}$, and the fixed sourcing and exporting costs $\{f_d^M, f_f^M, f_d^X, f_f^X\}$. This section concludes by defining the general equilibrium of this small open economy.

Definition 1 *Given the fixed number of entrants $\{N_d, N_f\}$ and the set of parameters Λ, the general equilibrium of the small open economy contains all manufacturers' optimal trade participation decisions $\{J_i^{M*}, J_i^{X*}\}$, the ownership-specific cutoff productivities $\{\underline{\varphi}_d^*, \underline{\varphi}_f^*\}$, the exchange rate ϵ^*, the aggregate expenditure at home E_h^*, and the aggregate price index at home P_h^* such that*

1. *$\{J_i^{M*}, J_i^{X*}\}$ maximize each manufacturer's profit in Equation (11).*

2. $\{\underline{\varphi}_d^*, \underline{\varphi}_f^*\}$ *satisfy the zero-profit conditions in Equation (13).*

3. ϵ^* *ensures that the trade balance condition in Equation (14) is satisfied.*

4. E_h^* *satisfies the budget constraint in Equation (15).*

5. P_h^* *ensures that the goods market clearing condition in Equation (16) is satisfied.*

5 Structural Estimation

In this section, I describe the estimation of the model parameters, which has two main parts. On the one hand, some parameters are estimated via country-level gravity regressions, including the relative sourcing and market potential abroad $\{\frac{\Theta_a^M}{\Theta_h^M}, \frac{\Theta_a^X}{\Theta_h^X}\}$, demand elasticity σ, and input trade elasticity θ. On the other hand, the rest of the parameters are jointly estimated via the simulated method of moments (SMM). The first part of the estimation follows Antràs et al. (2017) very closely, so I relegate the details to Appendix D and focus on the second part of the estimation. I first explain the parameterization of the productivity and fixed costs for domestic and foreign-owned manufacturers. The estimation procedure and results are then provided.

5.1 Parameterization

A manufacturer of ownership status $i \in \{d, f\}$ is assumed to draw its core productivity φ from a Pareto distribution with shape parameter κ. The manufacturer will not produce anything if the draw is below its ownership-specific productivity cutoff $\underline{\varphi}_i$ pinned down by the zero-profit conditions in Equation (13).[22] In addition, firms draw their sourcing and exporting fixed costs from two separate log-normal distributions. Specifically, the means of the log-normal distributions are parameterized as $\{\log \beta_{cons}^M + \log \beta_f^M \mathbb{1}(\text{Foreign-owned}), \log \beta_{cons}^X + \log \beta_f^X \mathbb{1}(\text{Foreign-owned})\}$. The additional coefficients for foreign-owned firms $\{\beta_f^M, \beta_f^X\}$ indicate their cost ratios relative to the costs of domestic firms. Last, the variances of the log-normal distributions are denoted by $\{\beta_{disp}^M, \beta_{disp}^X\}$.

[22] Although $\{\underline{\varphi}_d, \underline{\varphi}_f\}$ are actually endogenously determined in the model, their baseline values are estimated jointly with the shape parameter κ. Given the baseline estimates, the fixed operation costs $\{f_d^O, f_f^O\}$ can be backed out from Equation (13), which are the deep parameters of the model. In the counterfactual analysis in Section 6, $\{\underline{\varphi}_d, \underline{\varphi}_f\}$ will be resolved when the foreign demand shock is imposed.

5.2 Simulation and Estimation Procedure

The estimation distinguishes objects fitted to firm-level moments from the parameter governing the contribution of foreign-owned profits to domestic income. Conditional on the elasticities and relative sourcing and sales potentials estimated in Appendix D, the vector of objects fitted to firm-level moments is

$$\begin{aligned}\boldsymbol{\delta} = \big(&\gamma, \underline{\varphi}_d, \underline{\varphi}_f, \kappa, B_h, \beta^M_{cons}, \beta^M_f, \beta^M_{disp}, \\ &\beta^X_{cons}, \beta^X_f, \beta^X_{disp}\big)'.\end{aligned} \tag{18}$$

The productivity cutoffs and domestic demand term are baseline equilibrium objects. They are allowed to adjust in the counterfactual equilibrium, whereas the structural parameters governing technology and fixed costs remain fixed.

For each candidate parameter vector, I simulate 66,300 domestic firms and 6,400 foreign-owned firms, approximately matching the number of active manufacturers in the 2017 VES data. Productivity draws follow ownership-specific Pareto distributions with a common shape parameter. Sourcing and exporting fixed costs follow the distributions described in Section 5.1. Each simulated firm chooses among four alternatives: serving only the domestic market, importing only, exporting only, or both importing and exporting. I then construct moments from the resulting sales, wage bills, and trade participation decisions. Simulation draws and integration weights are held fixed during each estimation run.

Let m_j^{data} denote empirical moment j and $m_j^{\text{sim}}(\boldsymbol{\delta})$ its simulated counterpart. The criterion for the fourteen firm-level moments is

$$Q_{\text{firm}}(\boldsymbol{\delta}) = \sum_{j=1}^{14} \left| \frac{m_j^{\text{sim}}(\boldsymbol{\delta}) - m_j^{\text{data}}}{\left| m_j^{\text{data}} \right|} \right|. \tag{19}$$

Thus, the estimation minimizes absolute relative moment discrepancies. The normalization makes moments measured on different scales comparable, although moments with small empirical values receive greater weight per unit of absolute discrepancy.

The numerical implementation also includes an aggregate income-ratio target used to calibrate λ_f:

$$\left(\widehat{\boldsymbol{\delta}}, \widehat{\lambda}_f\right) \in \underset{\boldsymbol{\delta}\in\mathcal{D},\, \lambda_f\in[0,1]}{\arg\min} \left\{ Q_{\text{firm}}(\boldsymbol{\delta}) + \left| \frac{r^{\text{sim}}(\boldsymbol{\delta}, \lambda_f) - r_0}{r_0} \right| \right\}, \tag{20}$$

where r_0 is the empirical GNI-to-GDP ratio in 2017, and $\mathcal{D}$ denotes the admissible parameter region. The optimization uses bounded simulated annealing from multiple starting values, followed by bounded local refinement. Although λ_f is included in the numerical search, its information source differs from that of the firm-level parameters: it enters only the aggregate income-ratio moment.

5.3 Firm-Level Moments and Identification

Fourteen firm-level moments discipline the simulated firm outcomes. The following discussion describes the main information supplied by each group of moments; it does not assign each parameter exclusively to a particular moment. The complete list of moments is in Table 8.

The first set of moments concerns the estimation of productivity cutoffs and shape parameter $\{\underline{\varphi}_d, \underline{\varphi}_f, \kappa\}$. When productivity is Pareto distributed, the log share of firms with productivity larger than each productivity percentile is proportional to the log percentile.[23] Since the firm productivity is unobserved from the data, I instead take advantage of the distribution of sales in the data and implement the following regression specification for each sales percentile by ownership:

$$\log \underbrace{Pr(\text{Sales} \geq x_{pi})}_{\text{Share of firms with sales more than percentile } x_{pi}} = a_{0i} + a_{1i} \log x_{pi} + \epsilon_{pi} \; \forall i \in \{d, f\}, \tag{21}$$

where x_{pi} is p–th percentile of sales for firms of ownership i in the data. The estimates of $\{a_{0d}, a_{0f}\}$ and $\{a_{1d}, a_{1f}\}$ are then used as data moments. I conduct the same procedure with the simulated data and obtain the model moments.

The second set of moments deals with the fixed cost parameters $\{\beta^M_{cons}, \beta^M_f, \beta^M_{disp}, \beta^X_{cons}, \beta^X_f, \beta^X_{disp}\}$. To identify the average magnitude of sourcing and exporting fixed costs by firm ownership, I use the average shares of importers and exporters for domestic and foreign-owned firms in the 2017 VES data. Conditional on firm size, the discrepancy in trade participation rates between domestic and foreign-owned firms identifies the average fixed costs. Then, to identify the dispersion of these fixed costs, I pick the shares of importers and exporters among domestic and foreign-owned firms with sales below the median in the 2017 VES data. The intuition for the latter moments is that an increase in dispersion will lead to more importers and exporters among firms below the median sales than among those above the median when fixed cost draws are i.i.d. distributed. There are eight data moments used.

[23] If φ is Pareto distributed with scale $\underline{\varphi}_i$ and shape κ, then $\log Pr(\varphi \geq x) = \kappa \log \underline{\varphi}_i - \kappa \log x$.

The third set of moments takes care of the labor expenditure share γ and manufacturing residual demand at home B_h. Intuitively, I use the average wage bill over the sales ratio from the 2017 VES data as the moment to identify the labor expenditure share in the model. For the latter parameter, I adopt the share of firms with annual domestic sales less than the median (95K USD) in the 2017 VES data. This moment from the data is simply 0.5 by construction, but in the model, the domestic demand term that governs the magnitude of domestic sales will have to adjust to bring the simulated moment close to the data moment.

5.4 Calibration of Retained Profit Share

The retained foreign profit share, λ_f, is calibrated using Vietnam's GNI-to-GDP ratio in 2017, $r_0 = 0.93$. To describe the implemented calibration, let $\widetilde{W}(\boldsymbol{\delta})$ denote the total wage bill and let $\widetilde{\Pi}_d(\boldsymbol{\delta})$ and $\widetilde{\Pi}_f(\boldsymbol{\delta})$ denote weighted domestic and foreign-owned net profits. The calibration ratio is

$$r^{\text{sim}}(\boldsymbol{\delta}, \lambda_f) = \frac{\widetilde{W}(\boldsymbol{\delta}) + \widetilde{\Pi}_d(\boldsymbol{\delta}) + \lambda_f \widetilde{\Pi}_f(\boldsymbol{\delta})}{\widetilde{W}(\boldsymbol{\delta}) + \widetilde{\Pi}_d(\boldsymbol{\delta}) + \widetilde{\Pi}_f(\boldsymbol{\delta})}. \tag{22}$$

Define

$$s_f(\boldsymbol{\delta}) = \frac{\widetilde{\Pi}_f(\boldsymbol{\delta})}{\widetilde{W}(\boldsymbol{\delta}) + \widetilde{\Pi}_d(\boldsymbol{\delta}) + \widetilde{\Pi}_f(\boldsymbol{\delta})}. \tag{23}$$

For a given $\boldsymbol{\delta}$ with $s_f > 0$, exact matching of the income-ratio target implies

$$\lambda_f(\boldsymbol{\delta}; r_0) = 1 - \frac{1 - r_0}{s_f(\boldsymbol{\delta})}. \tag{24}$$

This expression applies when the implied value lies in $[0, 1]$. Otherwise, the bounded calibration selects the nearest endpoint, and the target cannot be matched exactly at that parameter vector.

Whenever exact matching is feasible, the income-ratio discrepancy can be eliminated by adjusting λ_f, without changing any firm-level moment. In that case, the joint criterion is equivalent to fitting the firm-level moments and subsequently recovering λ_f from equation (24). The calibrated value is therefore conditional on the fitted foreign-profit share and the imposed income target.

The tildes distinguish these calibration quantities from the complete income aggregates used in the counterfactual equilibrium. Interpreting r_0 as national GNI relative to GDP requires consistent sector coverage and income accounting. In particular, profits attributable to foreign investors do not become domestic national income merely because they are rein-

vested locally. Accordingly, λ_f is treated here as a model-specific income-allocation calibration, rather than a directly measured share of earnings retained by foreign-owned firms.

5.5 Bootstrap Uncertainty

I assess the sensitivity of the fitted objects to the firm sample using 25 bootstrap replications. Firms are resampled within strata that preserve ownership counts and membership in the samples used to construct the aggregate and sales-distribution moments. Firms shared by the two samples receive the same resampling multiplicity. For each replication, I reconstruct the empirical moments and repeat the numerical estimation and calibration.

The aggregate income-ratio target remains fixed at $r_0 = 0.93$. The elasticities and relative sourcing and sales potentials estimated outside this procedure are also held fixed. The bootstrap therefore does not incorporate uncertainty in these inputs.

Let $\widehat{q}$ denote a component of the baseline fitted vector and $\widehat{q}^{*(b)}$ its value in bootstrap replication b. The current implementation summarizes dispersion using the root mean square error (RMSE):[24]

$$\mathrm{RMSE}(\widehat{q}) = \left[\frac{1}{B}\sum_{b=1}^{B}\left(\widehat{q}^{*(b)} - \widehat{q}\right)^2\right]^{1/2}, \qquad B = 25. \tag{25}$$

5.6 Estimation and Calibration Results

The estimation results are summarized in Table 5. Regarding the productivity parameters, the productivity cutoff of foreign-owned firms is higher than that of domestic firms, reflecting the former's higher productivity. Then, note that the estimates of the fixed cost parameters reveal substantial differences between domestic and foreign-owned firms. In particular, β_f^M and β_f^X indicate that the sourcing and exporting costs for foreign-owned firms are only approximately 11 percent and 21 percent of the domestic firms' costs, respectively.[25] Foreign-owned firms retain 5.5% of their net profits in Vietnam according to my calibration.

The median sourcing and exporting fixed costs are shown in Table 6. Consistent with our parameter estimates, foreign-owned firms have significantly lower fixed costs, ranging from

[24] This statistic measures deviations from the baseline estimate and includes any shift of the bootstrap mean away from that estimate. It differs from the conventional bootstrap standard error, which is the standard deviation of the bootstrap estimates around their own mean.

[25] Because the fixed costs are assumed to be drawn from a log-normal distribution, their means are $\exp(\log \beta_{cons} + \mathbb{1}(i = f) \log \beta_f + \frac{\beta_{disp}^2}{2})$.

90K–341K USD, while the median costs for domestic firms range from 466K–3.18M USD. In addition, the fixed operation costs can be inferred from the lowest simulated profits (gross of operation costs) by ownership based on the zero-profit conditions in Equation (13). The result in Table 7 confirms that foreign-owned firms in Vietnam face higher operation costs to remain active in production activities, which in turn leads to their higher productivity than domestic firms on average.

5.7 Model Fit

The key moments of interest are the trade participation rate by ownership status. In Figure 4, I show that the SMM does a great job of matching the participation rate from the 2017 VES data. The remaining moment fits are provided in Table 8.

Figure 4: Moment Fit: Pre-Shock Trade Participation by Firm Ownership Status

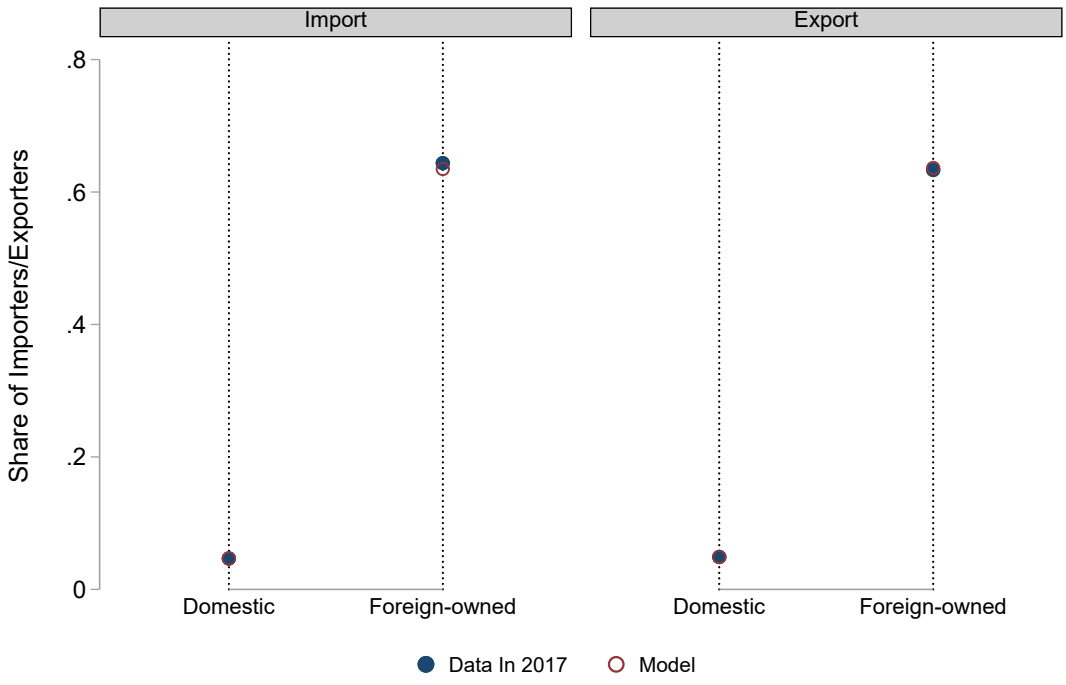


Table 5: Summary of Model Objects and Baseline Estimates

Object	Description	Value (S.E.)	Section
Estimation via linear regressions			
Θ_a^M/Θ_h^M	Relative sourcing potential abroad	1.611 (–)	D.1
Θ_a^X/Θ_h^X	Relative market potential abroad	1.526 (–)	D.1
σ	Demand elasticity	5.205 (0.440)	D.2
θ	Input trade elasticity	1.334 (0.363)	D.2
Estimation via simulated method of moments (SMM)			
$\underline{\varphi}_d$	Productivity cutoff of domestic firms	0.791 (0.044)	5
$\underline{\varphi}_f$	Productivity cutoff of foreign-owned firms	0.898 (0.045)	5
κ	Productivity dispersion	4.205 (0.136)	5
β_{cons}^M	Domestic sourcing fixed-cost scale	3.187 (0.785)	5
β_f^M	Foreign/domestic sourcing fixed-cost ratio	0.105 (0.031)	5
β_{disp}^M	Dispersion of log sourcing fixed costs	0.310 (0.163)	5
β_{cons}^X	Domestic exporting fixed-cost scale	0.466 (0.187)	5
β_f^X	Foreign/domestic exporting fixed-cost ratio	0.210 (0.066)	5
β_{disp}^X	Dispersion of log exporting fixed costs	0.459 (0.196)	5
γ	Labor input share	0.338 (0.067)	5
B_h	Residual demand at home	0.024 (0.005)	5
Calibration			
λ_f	Locally retained share of foreign-owned profits	0.055 (–)	5

Note: The regression parentheses contain standard errors, while SMM parentheses contain bootstrapped root mean square errors.

Table 6: Median Fixed Costs by Firm Ownership Status

	Median Value (Thousand USD)
f^M for domestic firms	3,177
f^M for foreign-owned firms	341
f^M for all firms	3,070
f^X for domestic firms	466
f^X for foreign-owned firms	90
f^X for all firms	439

Table 7: Operation Costs by Firm Ownership Status

	Value (USD)
f^O for domestic firms	8,771
f^O for foreign firms	14,949

Table 8: Complete List of Moments

Moment in 2017	Data	Model
Moments for productivity		
$\log \Pr(\text{Sales} \geq x_{pi}) = a_{0i} + a_{1i} \log x_{pi} + \epsilon_{pi} \; \forall i \in \{d, f\}$		
Intercept for domestic firms	-1.852	-2.211
Coefficient for $\log x_{pd}$	-0.334	-0.599
Intercept for foreign-owned firms	-0.847	-0.847
Coefficient for $\log x_{pf}$	-0.309	-0.416
Moments for fixed costs		
Share of domestic importers	0.047	0.047
Share of foreign importers	0.644	0.634
Share of domestic exporters	0.049	0.049
Share of foreign exporters	0.633	0.636
Share of domestic importers w/ less than median sales	0.002	0.000
Share of foreign importers w/ less than median sales	0.004	0.000
Share of domestic exporters w/ less than median sales	0.003	0.000
Share of foreign exporters w/ less than median sales	0.004	0.000
Moments for labor expenditure and manufacturing demand		
Share of firms w/ less than median domestic sales	0.500	0.500
Average wage bill over sales	0.338	0.273

6 Counterfactual Analysis of Foreign Demand Shocks

With the model parameters estimated from the previous section, I now conduct a counterfactual analysis of a foreign demand shock to the Vietnamese economy, generating trade responses from Vietnamese manufacturers of similar magnitudes to those in the actual data during the US–China trade war in 2017–2019. There are two main objectives of this exercise. The first is to understand the welfare implications of the shock when foreign-owned firms retain different levels of profits in Vietnam. The second is to understand the extent to which differences in the trade response between domestic and foreign-owned firms can be explained by their differences in productivity and fixed costs of sourcing and exporting.

6.1 Baseline Predictions

The baseline estimates of my model in Section 5 reflect the underlying primitives in 2017. To simulate the response from Vietnamese manufacturers to the tariff shock of the US–China trade war in 2018–2019, I increase the market potential abroad Θ_a^X such that the average export growth generated from the model is approximately 11.8%, same as the estimated export growth induced by the US trade war tariffs.[26] Specifically, the market potential abroad is raised by 49.9%, and the residual demand at home, exchange rate, aggregate price, aggregate expenditure, and number of active firms are resolved such that the zero-profit conditions in Equation (13), the trade balance condition in Equation (14), and the goods market clearing condition in Equation (16) are all satisfied under the new equilibrium.[27]

The percentage changes of key model moments in response to the foreign demand shock are summarized in Table 9. First, Vietnam's nominal exchange rate appreciates by 9% to offset the trade surplus induced by the positive demand shock. Due to heightened competition at home, Vietnam's aggregate price index is lowered by 6.2%. Overall, consumer welfare as measured by real expenditure increases by 7.95%. In Panel C, the welfare change is decomposed into labor income, domestic profits, and retained foreign profits, with the first component accounting for two-thirds of the increase (5.27 percentage points, or 66% of the overall change).

The changes in firm production outcomes over the initial distribution are demonstrated in Figure 5. Consistent with the changes in model moments after the shock, harsher competition

[26] The details of how I estimate the average export growth induced by the trade war tariffs are provided in Appendix E.1.

[27] The complete algorithm is provided in Appendix E.2.

and rising wages at home lead to negative growth in sales for most firms except the largest, which take advantage of the increasing demand abroad and gain in both local and export sales. Most firms also decrease their employment, resulting in negative wage bill growth.

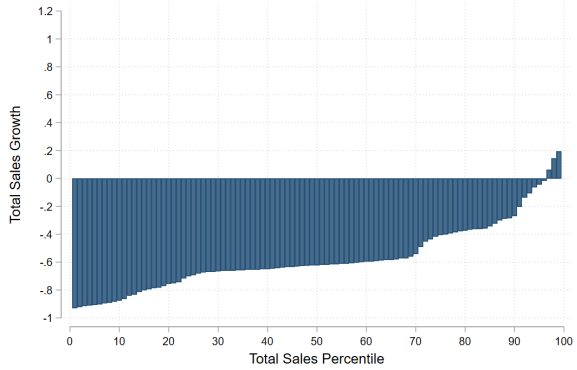
Table 9: Percentage Change of Model Objects to the Foreign Demand Shock

Model Object	Notation	Change
A. Foreign-demand shock (%)		
Foreign market potential (foreign currency)	Θ_a^X	49.90
B. Equilibrium outcomes (%)		
Nominal exchange rate	ϵ	-9.02
Aggregate price index	P_h	-6.20
Real expenditure	E_h/P_h	7.95
C. Welfare components (percentage points)		
Labor income	$w_h L_h/P_h$	5.27
Domestic profits	$N_d \int_{\underline{\varphi}_d}^{\infty} \pi_d(\varphi)\, dG(\varphi)/P_h$	2.65
Retained foreign profits	$\lambda_f N_f \int_{\underline{\varphi}_f}^{\infty} \pi_f(\varphi)\, dG(\varphi)/P_h$	0.03

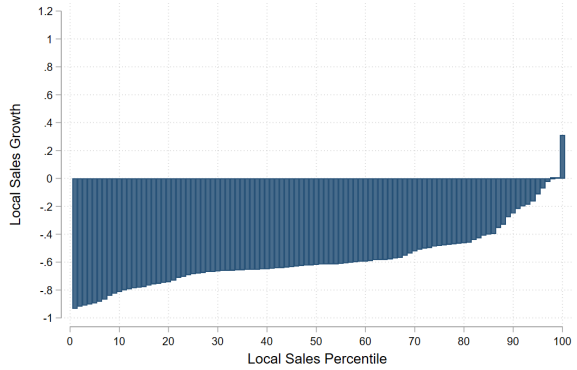
Note: Panel A reports the imposed foreign-demand shock. Panel B reports percentage changes in equilibrium outcomes. The nominal exchange rate is home currency per unit of foreign currency; a decline denotes appreciation. Panel C decomposes welfare growth into income contributions, measured as $100 \times \Delta X/(E_{h0}/P_{h0})$, where X is the real income component shown. The three contributions sum to real-expenditure growth in Panel B before rounding.

Figure 5: Change in Sales and Wage Bill of Manufacturers

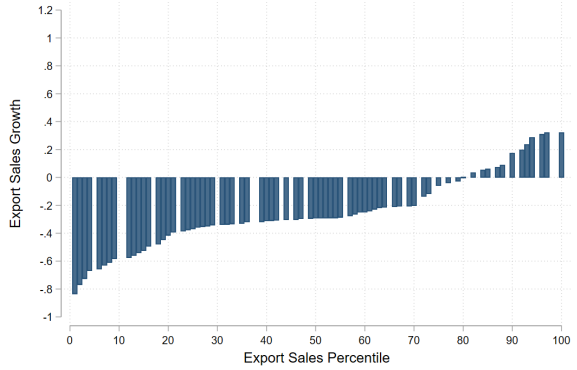
(a) Total Sales



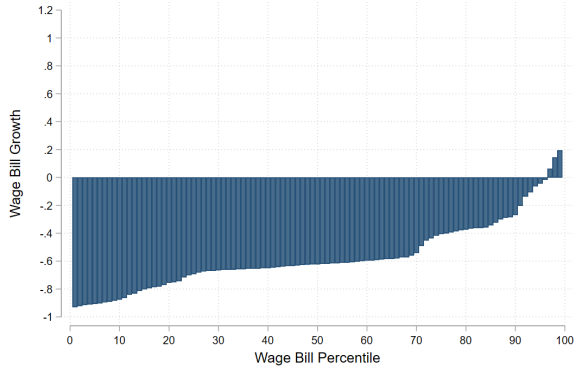
(b) Local Sales



(c) Export Sales



(d) Total Wage Bill



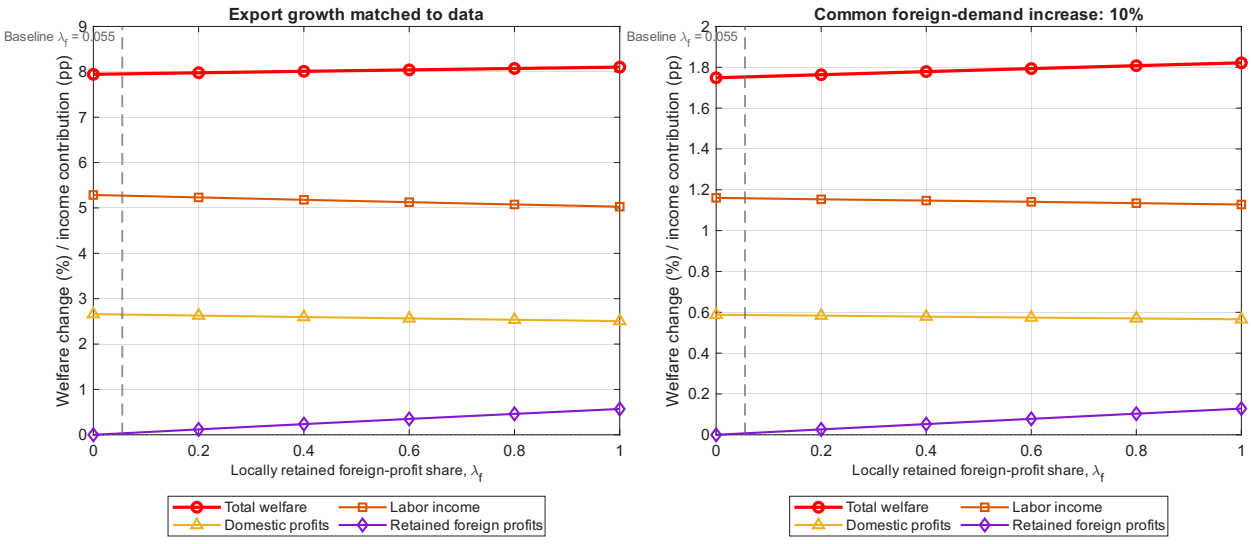
6.2 Welfare Effects of Foreign Profit Repatriation

The baseline calibration implies that only 5.5% of foreign-owned firms' profits are retained locally. Following Equation (15), I decompose real expenditure, E_h/P_h, into real labor income, domestic net profits, and locally retained foreign net profits. Each component's contribution is measured relative to baseline real expenditure, so the contributions sum to the percentage change in welfare. At the calibrated value of λ_f, the foreign-demand shock raises real expenditure by approximately 7.95%. Real labor income contributes 5.27 percentage points, domestic profits contribute 2.65 percentage points, and retained foreign profits contribute 0.03 percentage points. With the home wage normalized to one and aggregate labor supply fixed, the increase in real labor income reflects the decline in the aggregate price index.

To examine how profit repatriation affects these gains, the left panel of Figure 6 varies λ_f from zero to one, recalibrating the foreign-demand shock at each value to match the estimated 11.8% export growth. Welfare gains increase modestly, from 7.94% under full repatriation to 8.10% under full local retention. Over this range, the labor-income contribution falls from 5.29 to 5.03 percentage points, and the domestic-profit contribution declines from 2.66 to 2.50 percentage points. These declines are more than offset by the retained foreign-profit contribution, which rises from zero to 0.57 percentage points.

Because the shock required to match export growth varies with λ_f, the right panel holds the foreign-demand increase fixed at 10%. Welfare gains again rise with local profit retention, from 1.75% at $\lambda_f = 0$ to 1.82% at $\lambda_f = 1$. The retained foreign-profit contribution increases from zero to 0.13 percentage points, outweighing smaller declines in the labor-income and domestic-profit contributions. Thus, in both exercises, reducing profit repatriation modestly increases the welfare gain while shifting its composition toward retained foreign profits. Labor income remains the largest contributor throughout.

Figure 6: Decomposition of Welfare Change by Retained Foreign Profit Share



Note: Welfare is real expenditure, E_h/P_h. Total welfare reports its percentage change from the no-shock baseline at each λ_f. Income contributions are $100 \times \Delta X/(E_{h0}/P_{h0})$ percentage points and sum to total welfare, where X is each real income component. Profits are net of operating costs; foreign profits are multiplied by the locally retained share, λ_f. Left: foreign demand is calibrated at each λ_f to match 11.789% export growth. Right: a common 10% foreign-demand increase. Dashed vertical lines mark the calibrated baseline, $\lambda_f = 0.055$. Markers show the simulated values; lines connect them.

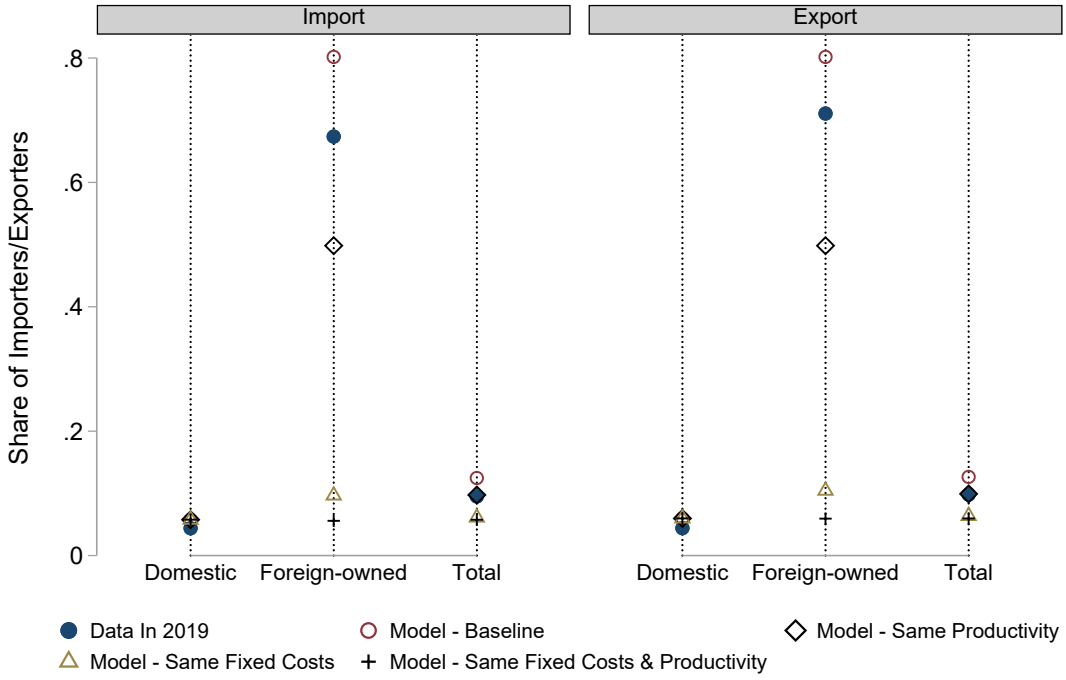
6.3 Decomposing the Roles of Productivity and Fixed Costs

Figure 7 compares the model's trade participation predictions following the foreign-demand shock with the 2019 data. Although the shock is calibrated to match aggregate export growth rather than participation rates, the model reproduces the substantial ownership gaps in trade participation while overpredicting participation levels. Predicted domestic import and export participation rates are approximately 5.8% and 6.0%, respectively, compared with 4.4% for both in the data. For foreign-owned firms, the model predicts import and export participation rates of 80.2%, exceeding the observed rates of 67.4% and 71.1%, respectively. Aggregate import and export participation rates are 12.5% and 12.6% in the model, compared with 9.5% and 9.8% in the data. Substantial gaps between domestic and foreign-owned firms persist in both the model and the data.

To assess the sources of these ownership gaps, I conduct two decomposition exercises, holding the counterfactual equilibrium environment and active-firm masses fixed. First, I set the productivity cutoff of foreign-owned firms equal to that of domestic firms while preserving ownership differences in trade fixed costs. Foreign-owned firms' import and export participation rates both fall to 49.8%. The remaining ownership gaps amount to approximately 59.2% of the baseline counterfactual import gap and 59.1% of the export gap.

Second, I equalize trade fixed-cost parameters across ownership groups while preserving productivity differences. Foreign-owned firms' import and export participation rates decline to 9.7% and 10.4%, respectively. The remaining gaps represent 5.2% of the baseline counterfactual import gap and 6.0% of the export gap. Equalizing both productivity and fixed costs nearly eliminates the ownership gaps: foreign-owned firms' import and export participation rates fall to 5.6% and 5.9%, respectively, compared with 5.8% and 6.0% for domestic firms. These results indicate that fixed-cost differences sustain a substantially larger participation gap on their own than productivity differences do. Nevertheless, both margins matter: their effects interact through firms' discrete trade decisions, so the remaining-gap ratios are not additive contributions.

Figure 7: Post-Shock Trade Participation Under Different Scenarios



Note: This figure compares 2019 import and export participation rates with model predictions following a foreign-demand shock calibrated to match aggregate export growth of 11.8%. "Model - Baseline" retains the estimated ownership differences in productivity and trade fixed costs. "Model - Same Productivity" sets foreign-owned firms' productivity cutoff equal to that of domestic firms; "Model - Same Fixed Costs" equalizes import and export fixed-cost parameters across ownership groups. The final specification imposes both restrictions. All decomposition exercises hold the counterfactual equilibrium environment and active-firm masses fixed.

7 Conclusion

The US-China trade war in 2018–2019 presents challenges but also opportunities for the bystander countries in the developing world. To the author's best knowledge, this paper is the first one to quantify the welfare gains of the trade war on a third country in a general equilibrium model. The role of foreign-owned manufacturers is highlighted, as they are often the largest and most productive firms in developing countries despite the concern of profit shifting and tax evasion. This paper takes the role of foreign ownership and profit repatriation seriously and studies its implications for trade participation, production responses, and real expenditure in the face of foreign demand shocks.

Focusing on Vietnam, I first show empirically that foreign-owned firms participate in trade much more than their domestic counterparts while paying less income taxes even conditional on firm size. Utilizing the US–China trade war episode as a foreign demand shock to Vietnam, I provide evidence that (1) the shock induced stronger import responses in intermediate goods from China and export responses in final goods to the US and the rest of the world and that (2) foreign-owned manufacturers in Vietnam increase their employment and assets in response to the shock, while domestic counterparts contract in employment and sales.

Based on the empirical findings, I develop and estimate a quantitative model of trade participation with foreign ownership and profit repatriation. In the counterfactual exercise, a foreign demand shock is fed into the model, generating the same magnitude of export growth as the trade war tariffs estimated from the data. The welfare decomposition implies that the gains from the demand shock in Vietnam are predominantly driven by the increase in labor income, not net profits. The productivity differences and ownership-specific fixed costs both play an important role in driving the differential response in trade participation between domestic and foreign-owned firms. Whether the net profits of the foreign-owned firms are retained locally matters for real expenditure in Vietnam but only to a mild extent, as full retention of foreign profits actually weakens the force of resource reallocation and lowers the real wage gains.

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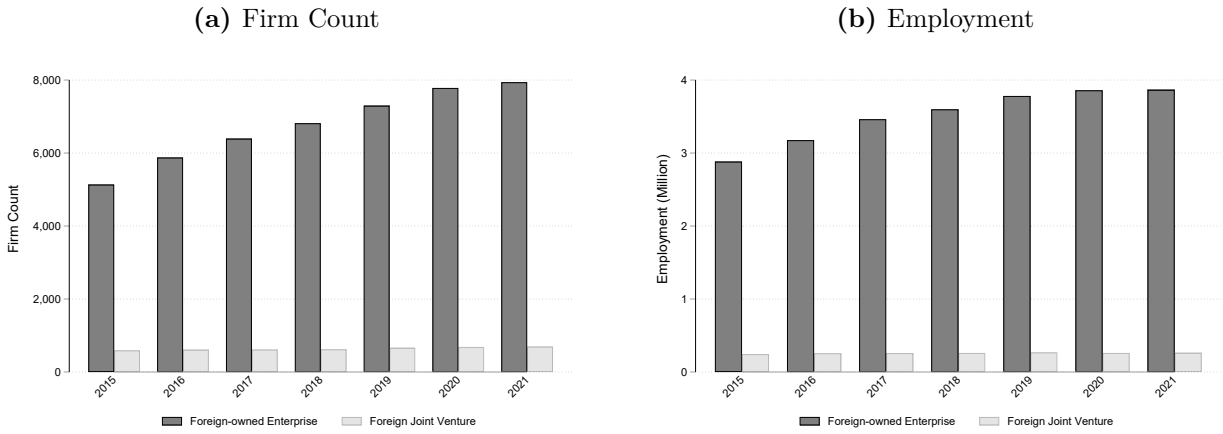
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A Additional Descriptive Facts

Fact 4. Foreign-owned manufacturers in Vietnam are predominantly greenfield investments.

In Figure A1, I show the firm count and employment for 100 percent foreign-owned manufacturers versus foreign joint ventures. It can be clearly seen that the foreign investments in Vietnam are predominantly greenfield FDI and that their count and employment were both increasing over time.

Figure A1: Vietnamese Manufacturers by Ownership Type in 2010–2021

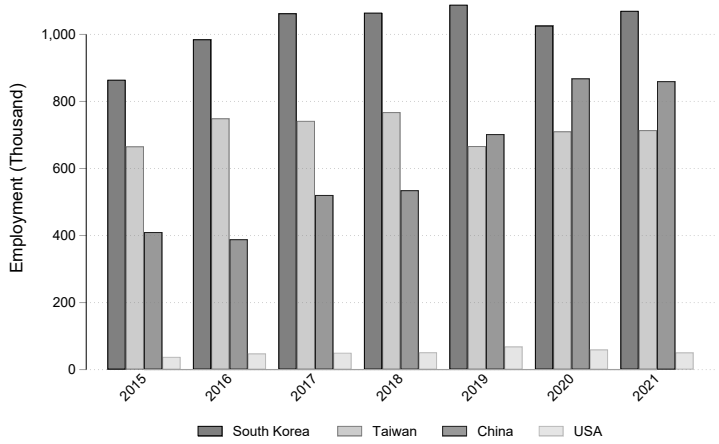


Note: Sample is Vietnamese manufacturers with positive employment, assets, and sales in the Vietnam Enterprise Survey from 2015 to 2021.

Fact 5. Foreign-owned manufacturers come mostly from nearby Asian countries.

Lastly, I show the employment of foreign-owned manufacturers by their origin in Figure A2. Foreign investments in Vietnam come mostly from nearby Asian countries. By the year 2019, the largest foreign investor in Vietnam was South Korea, followed by Taiwan and China. Notably, employment by Chinese manufacturers rose a lot after 2018 and became the second largest foreign investor by 2021.

Figure A2: Employment of Foreign-Owned Manufacturers by Origin (Top 3 + USA)



Note: The sample consists of Vietnamese manufacturers with positive employment, assets, and sales in the Vietnam Enterprise Survey from 2015 to 2021.

B Additional Results for the Reduced-Form Analysis

B.1 Trade Responses

Trade response by product type. I run the same specification on final goods and intermediate goods separately based on the Broad Economic Categories (BEC) of the United Nations Statistics Division. The estimates for final goods are in Table B1, and the estimates for intermediates are in Table B2. The growth of final goods exports from Vietnam to the US is particularly strong, as shown in column 2 of Table B1. On the other hand, the growth of intermediate imports from China to Vietnam is also large.

Table B1: Effect of US Trade War Tariffs on Vietnamese Trade Value (Final Goods)

(a) Export Value

	(1) Total Export	(2) Export to US	(3) Export to China	(4) Export to ROW
$\Delta \log(1+\tau_p^{US,CN})$	0.364	1.219	-1.524	0.239
	(0.480)	(0.753)	(1.114)	(0.477)
Observations	1355	853	643	1345
R^2	0.001	0.032	0.013	0.004

(b) Import Value

	(1) Total Import	(2) Import from US	(3) Import from China	(4) Import from ROW
$\Delta \log(1+\tau_p^{US,CN})$	0.742	-0.260	0.760	0.648
	(0.462)	(0.769)	(0.532)	(0.505)
Observations	1400	668	977	1368
R^2	0.007	0.008	0.017	0.003

Note: Standard errors are clustered at the product level. ROW stands for the rest of the world.

Table B2: Effect of US Trade War Tariffs on Vietnamese Trade Value (Intermediates)

(a) Export Value

	(1) Total Export	(2) Export to US	(3) Export to China	(4) Export to ROW
$\Delta \log(1+\tau_p^{US,CN})$	1.362	0.071	2.020	1.468
	(0.469)	(1.000)	(1.271)	(0.478)
Observations	2862	932	1021	2810
R^2	0.009	0.011	0.008	0.005

(b) Import Value

	(1) Total Import	(2) Import from US	(3) Import from China	(4) Import from ROW
$\Delta \log(1+\tau_p^{US,CN})$	0.035	0.435	0.795	-0.208
	(0.308)	(0.546)	(0.371)	(0.319)
Observations	3471	2001	2999	3417
R^2	0.001	0.000	0.004	0.009

Note: Standard errors are clustered at the product level. ROW stands for the rest of the world.

Alternative specification of the product-level analysis. To account for zero values for certain products in either 2017 or 2019, I employ the Davis–Haltiwanger growth rate measure for the product-level export and import values:

$$\frac{\Delta_{17,19} Y_p^{k,VN}}{\frac{1}{2}(Y_{p,17}^{k,VN} + Y_{p,19}^{k,VN})} = \beta_0 + \beta_1 \, \Delta_{17,19} \log(1+\tau_p^{US,CN}) + \beta_2 \, \Delta_{17,19} \log(1+\tau_p^{CN,US}) + \beta_3 \, \Delta_{17,19} \log(1+\tau_p^{US,VN}) + \beta_3 \, \frac{\Delta_{13,15} Y_p^{k,VN}}{\frac{1}{2}(Y_{p,13}^{k,VN} + Y_{p,15}^{k,VN})} + \epsilon_p, \tag{B1}$$

where $\Delta_{17,19} Y_p^{k,VN}$ is the difference in import/export values of an HS 6-digit product p between Vietnam and the country $k \in \{\text{All, US, China, ROW}\}$ from 2017 to 2019, $\Delta_{17,19} \log(1+\tau_p^{US,CN})$ is the change in the log US import tariff on exports of product p from China, $\Delta_{17,19} \log(1+\tau_p^{CN,US})$ is the change in the log Chinese import tariff on exports of product p from US, and $\Delta_{17,19} \log(1+\tau_p^{US,VN})$ is the change in the log US import tariff on exports of product p from Vietnam. Same as Equation (1), the growth rate from 2013 to 2015 is included to account for potential pre–trade war trends. As Table B3 shows, the strong trade responses to Trump tariffs, in particular exports to the US and imports from China, are also present under this alternative specification.

Table B3: Effect of US Trade War Tariffs on Vietnamese Trade Value (Davis–Haltiwanger Growth Rate)

(a) Export Value

	(1) Total Export	(2) Export to US	(3) Export to China	(4) Export to ROW
$\Delta \log(1+\tau_p^{US,CN})$	0.803	0.521	0.436	0.669
	(0.195)	(0.356)	(0.387)	(0.197)
Observations	4694	2442	2552	4651
R^2	0.009	0.005	0.002	0.007

(b) Import Value

	(1) Total Import	(2) Import from US	(3) Import from China	(4) Import from ROW
$\Delta \log(1+\tau_p^{US,CN})$	0.389	0.227	0.711	0.147
	(0.151)	(0.413)	(0.288)	(0.252)
Observations	5059	2969	3976	4785
R^2	0.003	0.001	0.002	0.003

Note: Standard errors are clustered at the product level. ROW stands for the rest of the world.

B.2 Production Responses

Construction of the sectoral exposure measure. The measure of sectoral exposure to the US–China trade war is constructed as a simple average of product-level tariffs. Specifically,

$$\Delta_{17,19} \log(1+\tau_{j(4)}^{US,CN}) = \sum_{p \in \Omega_{j(4)}} \frac{1}{\#\Omega_{j(4)}} \times \Delta_{17,19} \log(1+\tau_p^{US,CN}), \tag{B2}$$

$$\Delta_{17,19} \log(1+\tau_{j(4)}^{CN,US}) = \sum_{p \in \Omega_{j(4)}} \frac{1}{\#\Omega_{j(4)}} \times \Delta_{17,19} \log(1+\tau_p^{CN,US}), \tag{B3}$$

$$\Delta_{17,19} \log(1+\tau_{j(4)}^{US,VN}) = \sum_{p \in \Omega_{j(4)}} \frac{1}{\#\Omega_{j(4)}} \times \Delta_{17,19} \log(1+\tau_p^{US,VN}), \tag{B4}$$

where $j(4)$ stands for an ISIC 4-digit sector and p stands for a 6-digit HS product code. $\Omega_{j(4)}$ is the set of products belonging to sector $j(4)$.[28] The summary of the resulting exposure

[28] Ω_j is defined according to the HS–ISIC correspondence table from Jordan Industrial Observatory (2022).

measure is provided in Table B4.

Table B4: Summary of Sectoral Tariff Exposures in 2017–2019

	$\Delta_{17,19}\log(1+\tau_{j(4)}^{US,CN})$	$\Delta_{17,19}\log(1+\tau_{j(4)}^{CN,US})$	$\Delta_{17,19}\log(1+\tau_{j(4)}^{US,VN})$
Mean	0.141	0.113	0.000
Median	0.151	0.133	0.000
SD	0.070	0.051	0.004

Note: This table shows average tariffs at the ISIC 4-digit level.

Results for incumbent firms. I run the same specification as Equation (2) on manufacturers established in Vietnam before 2018. The DIDID result in Table B5 shows that an average change in US import tariffs reduces domestic firms' employment by 3% while increasing foreign-owned firms' employment by 8%. On the other hand, the tariff shock raises foreign firms' assets by 5%, and sales fall by 16% for domestic firms.

Table B5: Effect of Trade War Tariffs on Employment and Sales (Incumbents)

	(1) log Employment	(2) log Assets	(3) log Sales
$\Delta \log(1+\tau_{j(4)}^{US,CN}) \times Post_t$	-0.224	-0.022	-1.154
	(0.112)	(0.107)	(0.162)
$\Delta \log(1+\tau_{j(4)}^{US,CN}) \times Post_t \times Foreign_i$	0.775	0.408	0.888
	(0.197)	(0.162)	(0.278)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes
Foreign × Year FE	Yes	Yes	Yes
Total tariff effect: Foreign	0.551	0.386	-0.267
	(0.200)	(0.164)	(0.280)
Observations	371506	371506	371506
R-squared	0.894	0.918	0.888

Note: The sample includes Vietnamese manufacturers first appearing in the Vietnam Enterprise Survey before 2018 with positive employment, assets, and sales. Monetary variables are winsorized at the 1st and 99th percentiles. Regressions include firm, year, two-digit industry-by-year, and foreign-status-by-year fixed effects, and control for changes in Chinese tariffs on U.S. goods and U.S. tariffs on Vietnamese goods, interacted with foreign status and a post-2017 indicator. Standard errors are clustered at the firm level.

Alternative tariff exposure. An alternative sectoral exposure to the US–China trade war is constructed as a weighted average of product-level tariffs, with the weights given by the US export market shares of each Vietnamese product for a sector. Specifically,

$$\Delta_{17,19} \log(1+\tau_{w,j(4)}^{US,CN}) = \sum_{p \in \Omega_{j(4)}} \frac{X_{p,17}^{US,VN}}{X_{j(4),17}^{US,VN}} \times \Delta_{17,19} \log(1+\tau_{p}^{US,CN}), \tag{B5}$$

$$\Delta_{17,19} \log(1+\tau_{w,j(4)}^{CN,US}) = \sum_{p \in \Omega_{j(4)}} \frac{X_{p,17}^{US,VN}}{X_{j(4),17}^{US,VN}} \times \Delta_{17,19} \log(1+\tau_{p}^{CN,US}), \tag{B6}$$

$$\Delta_{17,19} \log(1+\tau_{w,j(4)}^{US,VN}) = \sum_{p \in \Omega_{j(4)}} \frac{X_{p,17}^{US,VN}}{X_{j(4),17}^{US,VN}} \times \Delta_{17,19} \log(1+\tau_{p}^{US,VN}), \tag{B7}$$

where $j(4)$ stands for an ISIC 4-digit sector and p stands for a 6-digit HS product code. $\Omega_{j(4)}$ is the set of products belonging to sector $j(4)$. The summary of the resulting exposure measure is provided in Table B6.

Table B6: Summary of Weighted Tariff Exposures in 2017–2019

	$\Delta_{17,19}\log(1+\tau^{US,CN}_{w,j(4)})$	$\Delta_{17,19}\log(1+\tau^{CN,US}_{w,j(4)})$	$\Delta_{17,19}\log(1+\tau^{US,VN}_{w,j(4)})$
Mean	0.152	0.129	0.003
Median	0.193	0.151	0.000
SD	0.080	0.071	0.022

Note: This table summarizes weighted tariffs at the ISIC 4-digit level.

Table B7 shows the DIDID estimates using the weighted tariff exposure as the treatment variable. Despite less precision in the estimates, the overall patterns are the same: foreign-owned firms in more exposed sectors increase their employment, while domestic firms contract in sales.

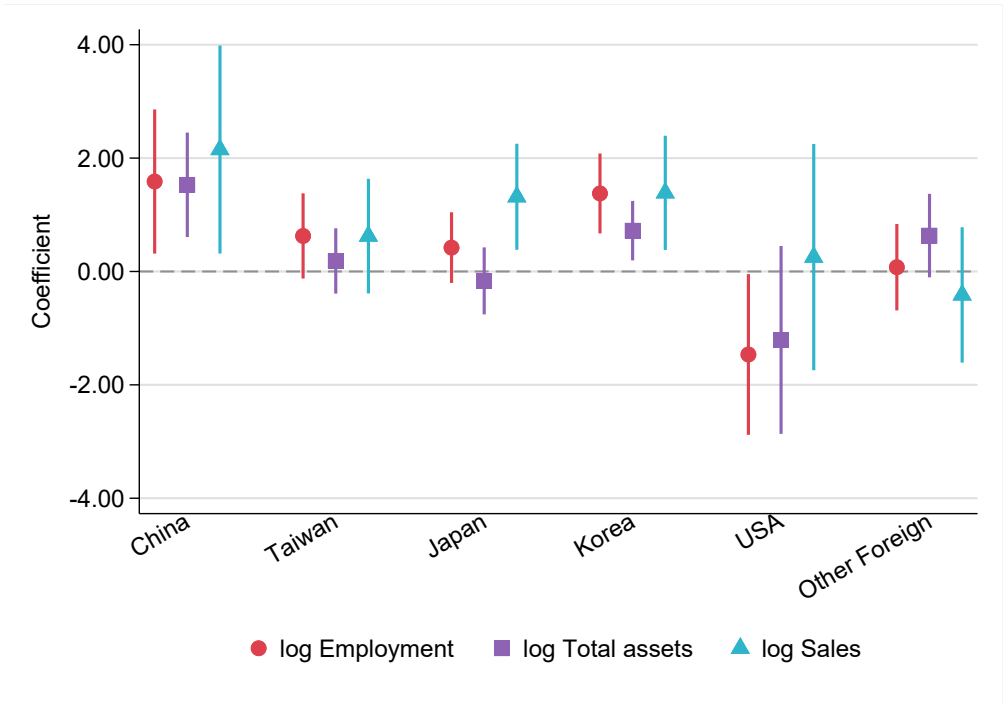
Table B7: Effect of Trade War Tariffs on Employment and Sales (Weighted Tariffs)

	(1) log Employment	(2) log Assets	(3) log Sales
$\Delta\log(1+\tau^{US,CN}_{w,j(4)}) \times Post_t$	-0.026	0.077	-0.260
	(0.080)	(0.075)	(0.118)
$\Delta\log(1+\tau^{US,CN}_{w,j(4)}) \times Post_t \times Foreign_i$	0.368	0.103	0.031
	(0.150)	(0.130)	(0.227)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry × Year FE	Yes	Yes	Yes
Foreign × Year FE	Yes	Yes	Yes
Total tariff effect: Foreign	0.342	0.180	-0.230
	(0.144)	(0.123)	(0.218)
Observations	457414	457414	457414
R-squared	0.894	0.921	0.882

Note: The sample consists of Vietnamese manufacturers present in the Vietnam Enterprise Survey from 2015 to 2021 with positive employment, assets, and sales. Monetary variables are winsorized at the 1st and 99th percentiles. Regressions include firm, year, two-digit industry-by-year, and foreign-status-by-year fixed effects, and control for changes in Chinese tariffs on U.S. goods and U.S. tariffs on Vietnamese goods, interacted with foreign status and a post-2017 indicator. Standard errors are clustered at the firm level.

Country heterogeneity. Figure B1 illustrates the DIDID estimates (β_1 in Equation (2)) for Vietnam's major foreign capital source countries. Chinese manufacturers exhibit the strongest response to Trump tariffs, followed by Korean and Japanese manufacturers.

Figure B1: Heterogeneity by Main Capital Source Country



C Proof of Proposition 1

Proof of part 1:

The goal is to show that $\pi(\varphi, J^M, J^X)$ has increasing differences in (φ, J^M) and (φ, J^X). Note that $\forall\, \varphi' > \varphi$,

$$\begin{aligned}
&\pi(\varphi', 1, J^X) - \pi(\varphi, 1, J^X) \\
&= (\varphi'^{\sigma-1} - \varphi^{\sigma-1}) w_h^{-\gamma(\sigma-1)} \beta^{-(\sigma-1)} [\Theta_h^M + \epsilon^{-\theta} \Theta_a^M]^{\frac{(1-\gamma)(\sigma-1)}{\theta}} [\Theta_h^X + J^X \epsilon^\sigma \Theta_a^X] \\
&\geq (\varphi'^{\sigma-1} - \varphi^{\sigma-1}) w_h^{-\gamma(\sigma-1)} \beta^{-(\sigma-1)} [\Theta_h^M]^{\frac{(1-\gamma)(\sigma-1)}{\theta}} [\Theta_h^X + J^X \epsilon^\sigma \Theta_a^X] \\
&= \pi(\varphi', 0, J^X) - \pi(\varphi, 0, J^X).
\end{aligned}$$

The first equality holds by Equation (11). The inequality holds because the sourcing potentials are strictly positive and $\frac{(1-\gamma)(\sigma-1)}{\theta} \geq 0$. The last equality again follows from Equation (11). Therefore, $\pi(\varphi, J^M, J^X)$ has increasing differences in (φ, J^M). Following the same exercise, it can be shown that $\pi(\varphi, J^M, J^X)$ also has increasing differences in (φ, J^X).

Proof of part 2:

Our goal is to show that $\pi(\varphi, J^M, J^X)$ has increasing differences in (J^M, J^X). Note that

$$\begin{aligned}
&\pi(\varphi, 1, 1) - \pi(\varphi, 1, 0) \\
&= \varphi^{\sigma-1} w_h^{-\gamma(\sigma-1)} \beta^{-(\sigma-1)} [\Theta_h^M + \epsilon^{-\theta} \Theta_a^M]^{\frac{(1-\gamma)(\sigma-1)}{\theta}} [\epsilon^\sigma \Theta_a^X] - w_h f_i^X \\
&\geq \varphi^{\sigma-1} w_h^{-\gamma(\sigma-1)} \beta^{-(\sigma-1)} [\Theta_h^M]^{\frac{(1-\gamma)(\sigma-1)}{\theta}} [\epsilon^\sigma \Theta_a^X] - w_h f_i^X \\
&= \pi(\varphi, 0, 1) - \pi(\varphi, 0, 0).
\end{aligned}$$

The first equality follows from Equation (11). The inequality holds because the sourcing potentials are strictly positive and $\frac{(1-\gamma)(\sigma-1)}{\theta} \geq 0$. The last equality again follows from Equation (11). Therefore, $\pi(\varphi, J^M, J^X)$ has increasing differences in (J^M, J^X).

D Additional Details of Structural Estimation

D.1 Estimation of Foreign Sourcing and Market Potentials

This section describes the estimation of relative sourcing and market potential abroad, i.e., $\{\frac{\Theta_a^M}{\Theta_h^M}, \frac{\Theta_a^X}{\Theta_h^X}\}$. Following the same approach as Antràs et al. (2017), I first estimate each foreign country's sourcing and market potential relative to Vietnam's and then average the figures by trade flows with Vietnam to obtain the "representative" sourcing and market potential abroad. The Vietnam Technology and Competitiveness Survey (TCS) asks manufacturers where their top three sourcing and exporting countries are and the associated firm-level import and export shares, which I use for the following estimation.

Note that every firm's import and export shares for each country are analogous to Equations (10) and (12):

$$\chi_{i,j}^M = \frac{(\tau_j^M w_j)^{-\theta} T_j}{\sum_l (\tau_l^M w_l)^{-\theta} T_l} = \frac{\Theta_j^M}{\sum_l \Theta_l^M}, \tag{D1}$$

$$\chi_{i,k}^X = \frac{(\tau_k^X)^{1-\sigma} B_k}{\sum_l (\tau_l^X)^{1-\sigma} B_l} = \frac{\Theta_k^X}{\sum_l \Theta_l^X}, \tag{D2}$$

where $\{\chi_{i,j}^M, \chi_{i,k}^X\}$ are firm i's import share from country j and export share to country k. Taking the log of the shares and subtracting the log shares in Vietnam, I can obtain the following:

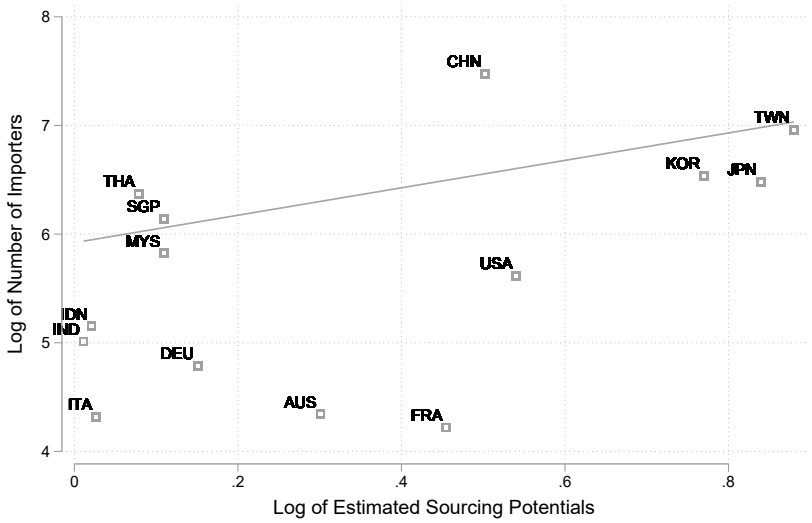
$$\log \chi_{i,j}^M - \log \chi_{i,h}^M = \log \underbrace{\Theta_j^M / \Theta_h^M}_{\text{Relative sourcing potential of country j}}, \tag{D3}$$

$$\log \chi_{i,k}^X - \log \chi_{i,h}^X = \log \underbrace{\Theta_k^X / \Theta_h^X}_{\text{Relative market potential of country k}}, \tag{D4}$$

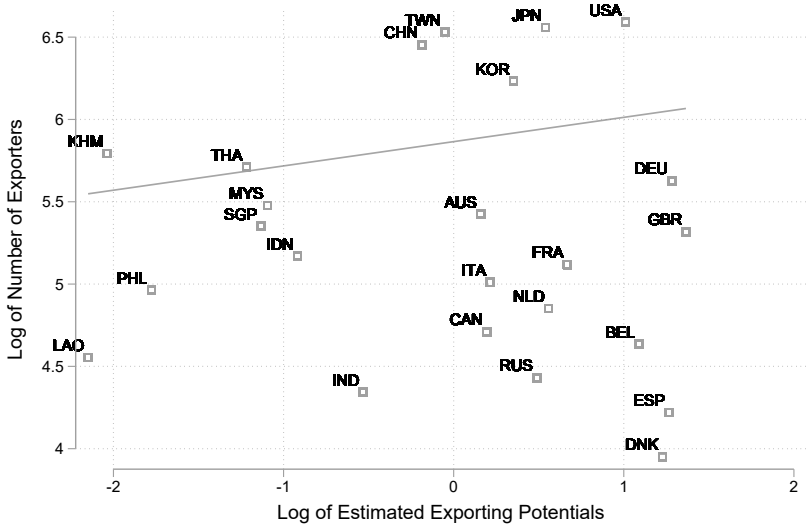
where $\{\chi_{i,h}^M, \chi_{i,h}^M\}$ are firm i's shares of domestic sourcing and sales. The relative sourcing and market potentials can then be estimated by regressing the log differences in shares on the country dummies. Figure D1 illustrates the estimates and associated log number of importers and exporters in the TCS data. Intuitively, the estimated sourcing potentials are high for Vietnam's main sourcing countries such as China, and the estimated market potentials are high for major exporting partners such as the US.

Figure D1: Estimates of Relative Sourcing and Market Potential by Country

(a) Sourcing Potential



(b) Market potential



Note: The sourcing and market potentials are estimated with the Vietnam Technology and Competitiveness Survey (TCS) from years 2009–2014. The estimates are relative to the potential for local sourcing and sales in Vietnam.

D.2 Estimation of Demand and Input Trade Elasticities

To estimate the demand elasticity σ, note that the market potential of country k is:

$$\Theta_k^X = (\tau_k^X)^{1-\sigma} B_k. \tag{D5}$$

Then, by taking the log of the estimated market potential (with Vietnam's potential normalized to one), σ can be estimated with the following regression specification:

$$\begin{aligned}\log \widehat{\Theta_k^X} =&(1-\sigma)\log \tau_k^X + \alpha_d \log \text{distance}_{hk} + \alpha_c \text{ control of corruption}_k + \alpha_g \log \text{GDP}_k \\ &+ \alpha_0 + \epsilon_k,\end{aligned} \tag{D6}$$

where variations in log average export tariffs to country k identify the demand elasticity. I further control for gravity variables such as bilateral distance, degree of corruption, and country GDP to account for potential omitted variable bias.[29] The estimation result is shown in Table D1. The estimates indicate an estimate of σ of approximately 5.2, implying an average markup of 24 percent.[30]

The strategy of estimating the input trade elasticity θ is analogous. Note that the sourcing potential of country j is:

$$\Theta_j^M = (\tau_j^M w_j)^{-\theta} T_j. \tag{D7}$$

Taking the log of the estimated sourcing potentials, I estimate θ with the following specification:

$$\begin{aligned}\log \widehat{\Theta_j^M} =& -\theta \log(\tau_j^M w_j) + \beta_d \log \text{distance}_{hj} + \beta_c \text{ control of corruption}_j + \beta_g \log \text{GDP}_j \\ &+ \beta_r \log \text{R\&D}_j + \beta_k \log \text{capital}_j + \beta_e \log \#\text{ establishments}_j + \beta_0 + \epsilon_j,\end{aligned} \tag{D8}$$

where variations in $\log(\tau_j^M w_j)$, the log multiple of average import tariffs and manufacturing monthly wage in country j,[31] identify the input trade elasticity. In addition to the standard gravity variables, I control for R&D expenditure, the capital–labor ratio, and the number of establishments in each foreign country j to account for the technology level term T_j.

[29] Details of the estimation and data sources are provided in Appendix D.3.

[30] The estimate of σ in Antràs et al. (2017) with the US data is 3.85, implying an average markup of 35 percent.

[31] Wages are adjusted by country-level human capital based on years of schooling from the Barro–Lee Education Attainment Dataset and the returns to education from Psacharopoulos and Patrinos (2018).

To further deal with unobserved variables that could correlate with both the manufacturing wage and the error term, I follow the same approach as in Antràs et al. (2017), using country population as an instrument. The results are shown in Table D2. The instrument seems to do a reasonably good job of correcting the selection bias associated with sourcing activities at the country level. I end up using $\theta = 1.334$.[32]

Table D1: Estimation of Demand Elasticity

	Log Exporting Potential		
$\log \tau_k^X$	-4.223	-4.324	-4.205
	(0.342)	(0.451)	(0.440)
log Distance		1.891	-2.978
		(5.297)	(1.219)
Control of Corruption		0.300	0.163
		(0.271)	(0.175)
log GDP			0.805
			(0.402)
Constant	0.222	-15.517	13.206
	(0.015)	(43.602)	(15.246)
Observations	142	142	142
R^2	0.919	0.920	0.921

Note: Sample is at the country-year level. Standard errors are clustered at the region level.

[32]The estimate of θ in Antràs et al. (2017) with the US data is 1.789.

Table D2: Estimation of Input Trade Elasticity

	Log Sourcing Potential	
	OLS	IV
$\log \tau_j^M w_j$	0.012	-1.334
	(0.056)	(0.363)
log Distance	0.085	0.111
	(0.056)	(0.110)
Control of Corruption	-0.249	0.499
	(0.044)	(0.227)
log GDP	-0.165	-0.809
	(0.112)	(0.233)
log R&D	0.219	0.637
	(0.033)	(0.214)
log Capital/Worker	0.009	0.391
	(0.135)	(0.154)
log # Establishment	-0.135	0.059
	(0.067)	(0.110)
Constant	1.418	7.289
	(1.106)	(3.190)
Observations	59	59
R^2	0.241	.

Note: Sample is at the country-year level. Standard errors are clustered at the region level.

D.3 Data Source for Estimation

Table D3: Data Source

Variable	Data Source
Bilateral trade flows	BACI, CEPII
Bilateral distance, language, contiguity	Gravity, CEPII
Control of corruption	WGI, World Bank
R&D expenditure	WDI, World Bank
Number of establishments	Industrial Statistics Database, UNIDO
Manufacturing wage	ILOSTAT, International Labour Organization
GDP	Penn World Table
Physical capital	Penn World Table
Years of schooling	Barro–Lee Educational Attainment Dataset
Mincer coefficients by country	Psacharopoulos and Patrinos (2018)

E Details of Counterfactual Analysis

E.1 Estimation of Average Export Response from the Trade War

Here I explain the estimation procedure to obtain the average export response of Vietnamese manufacturers induced by the US import tariffs for the counterfactual analysis. A regression specification similar to Equation (B1) is estimated for all final goods exports from Vietnam:

$$\frac{\Delta_{17,19} X_p^{VN}}{\frac{1}{2}(X_{p,17}^{VN} + X_{p,19}^{VN})} = \beta_0 + \beta_1 \, \Delta_{17,19} \log(1 + \tau_p^{US,CN}) + \epsilon_p, \tag{E1}$$

where p denotes HS 6-digit products, and X_p^{VN} is the total export value of product p from Vietnam. With the estimates of β_0 and β_1, the predicted export growth can be derived by rearranging terms and summing over all products:

$$g_X^{\text{data}} = \frac{\sum_{p=1}^{N} \widehat{\Delta_{17,19} X_p^{VN}}}{\sum_{p=1}^{N} X_{p,17}} = \frac{\sum_{p=1}^{N} \frac{1}{2}(X_{p,17}^{VN} + X_{p,19}^{VN}) \times \left[\widehat{\beta}_0 + \widehat{\beta}_1 \, \Delta_{17,19} \log(1 + \tau_p^{US,CN})\right]}{\sum_{p=1}^{N} X_{p,17}}. \tag{E2}$$

E.2 Algorithm

The parameters in the model include demand elasticity σ, input trade elasticity θ, labor expenditure share γ, sourcing and market potential abroad $\{\Theta_a^M, \Theta_a^X\}$, the productivity shape parameter κ, fixed-cost parameters $\{\beta_{cons}^M, \beta_f^M, \beta_{disp}^M, \beta_{cons}^X, \beta_f^X, \beta_{disp}^X\}$, fixed operation costs $\{f_d^O, f_f^O\}$, potential entrant masses $\{N_d, N_f\}$, and the locally retained share of foreign-owned profits λ_f.

The endogenous variables include trade participation decisions $\{J_i^M, J_i^X\}$, ownership-specific productivity cutoffs $\{\underline{\varphi}_d, \underline{\varphi}_f\}$, the exchange rate ϵ, aggregate expenditure at home E_h, and the aggregate price index at home P_h. Domestic sales potential B_h is linked to expenditure and the price index through equation (6).

I compute the counterfactual using fixed integration nodes, a joint solution of the equilibrium conditions for a given foreign-demand shock, and an outer scalar search that matches aggregate export growth. The procedure below describes the common-productivity-cutoff implementation. Potential entrant masses and structural parameters are held fixed throughout each exercise.

Units and notation. The home wage is the numeraire, $w_h = 1$, and aggregate labor supply L_h is fixed. Let $\underline{\varphi}_i$ denote the productivity cutoff of ownership group $i \in \{d, f\}$, and let $\eta = \epsilon/\epsilon_0$ denote the exchange rate relative to baseline. The exchange rate is home currency per unit of foreign currency, so $\eta > 1$ is a depreciation. A subscript zero denotes the baseline. Following equation (6), define

$$c_\sigma = \frac{1}{\sigma}\left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma}, \qquad B_h = c_\sigma E_h P_h^{\sigma-1}. \tag{E3}$$

Write the foreign demand shock as the level change $\Delta\Theta_a^X$ and define $s = 1 + \Delta\Theta_a^X/\Theta_{a0}^X$, the corresponding multiplier on exogenous foreign market potential. Following equation (11), define the sourcing and market potentials without exchange-rate factors:

$$\begin{aligned} \Theta_h^M &= \frac{T_h}{w_h^\theta}, \qquad \Theta_a^M = \frac{T_a}{(\tau^M w_a)^\theta}, \\ \Theta_h^X &= B_h, \quad \Theta_a^X(s) = s\Theta_{a0}^X, \qquad \Theta_{a0}^X = \frac{B_{a0}}{(\tau^X)^{\sigma-1}}. \end{aligned} \tag{E4}$$

For the positive-demand experiment, $s \geq 1$. Foreign sourcing potential Θ_a^M is held fixed, while foreign market potential changes only through s. The exchange rate enters variable profits separately through the brackets:

$$\Theta_h^M + J_i^M \epsilon^{-\theta}\Theta_a^M, \qquad B_h + J_i^X \epsilon^\sigma s\Theta_{a0}^X.$$

The factor ϵ^σ follows because foreign buyers pay $\tau^X p_h/\epsilon$ and export receipts are converted back into home currency.

For computation in baseline ratios, define the exchange-rate-adjusted coefficients C_M and C_X, which are distinct from the potentials:

$$\begin{aligned} C_M(\eta) &= C_{M0}\eta^{-\theta}, \quad C_{M0} = \epsilon_0^{-\theta}\Theta_a^M, \\ C_X(s,\eta) &= C_{X0}s\eta^\sigma, \quad C_{X0} = \epsilon_0^\sigma\Theta_{a0}^X. \end{aligned} \tag{E5}$$

Thus, using $\eta = \epsilon/\epsilon_0$ is equivalent to using the exchange-rate level in equation (11), provided the baseline factors are included in C_{M0} and C_{X0}. Under the baseline currency normalization $\epsilon_0 = 1$, these coefficients equal the corresponding baseline foreign potentials. Foreign prices, foreign wages and bilateral trade costs are otherwise held fixed throughout the experiment.

1. Initialize the integration grid and baseline. For each ownership group, construct a fixed set of productivity quantiles and paired import/export fixed-cost draws. Denote the resulting nodes by $\{u_{ij}, f_{ij}^M, f_{ij}^X, \omega_{ij}\}_{j=1}^{n_i}$, where $\sum_j \omega_{ij} = 1$. The implementation uses 100 stratified productivity nodes, crossed with 663 fixed-cost pairs for domestic firms and 64 pairs for foreign-owned firms, giving $n_d = 66{,}300$ and $n_f = 6{,}400$ integration nodes. The productivity strata have boundaries: [0, .25, .50, .65, .80, .90, .95, .98, .99, .999, 1], with ten draws per stratum. Fixed-cost pairs are constructed using permuted van der Corput draws transformed into the specified lognormal distributions. All underlying draws and weights are retained throughout the exercise.

With Pareto lower support $\varphi_{\min} = 0.5$ and shape parameter κ, conditional productivity and economic active-firm mass are

$$\varphi_{ij}(\underline{\varphi}_i) = \underline{\varphi}_i(1-u_{ij})^{-1/\kappa}, \qquad M_i = N_i\big[1 - G(\underline{\varphi}_i)\big] = N_i\left(\frac{\varphi_{\min}}{\underline{\varphi}_i}\right)^{\kappa}, \qquad \underline{\varphi}_i \geq \varphi_{\min}. \tag{E6}$$

The fixed entrant masses are calibrated as

$$N_i = \frac{M_{i0}}{\left(\varphi_{\min}/\underline{\varphi}_{i0}\right)^{\kappa}}. \tag{E7}$$

Economic masses M_i and N_i are continuous quantities, not array dimensions; they are not rounded when solving the model. An aggregate firm outcome is computed as $\sum_i M_i \sum_j \omega_{ij} y_{ij}$, with no additional survival factor inside the conditional average.

I use the following boundary-profit rule in the current implementation:

$$b_i(\underline{\varphi}_i; B_h, \eta, s) = \min_{\ell \in \mathcal{F}_i} \max_{(J^M, J^X) \in \{0,1\}^2} \left\{ v_i(\underline{\varphi}_i, J^M, J^X; B_h, \eta, s) - J^M f_{i\ell}^M - J^X f_{i\ell}^X \right\}, \tag{E8}$$

where $\mathcal{F}_i$ is the fixed set of cost pairs, and v_i denotes π_v in equation (11), including its common production constant β and the explicit exchange-rate factors. Baseline operation costs are calibrated from $f_i^O = b_i(\underline{\varphi}_{i0}; B_{h0}, 1, 1)$ and are then fixed. This rule evaluates productivity at the cutoff itself, rather than at the lowest interior integration node.[33]

Using baseline firm outcomes, infer L_h from total labor expenditure, including assembly,

[33] Equation (E8) is an explicit operational rule for the current common-cutoff implementation. With heterogeneous trade costs known before the operation decision, entry generally depends jointly on productivity and those costs. If trade costs are learned after an irreversible operation decision, a common cutoff would instead be based on expected profit. The minimum-over-costs rule is not an expected-profit free-entry condition; the entry timing must be specified before giving it a structural interpretation.

local-input production and fixed-cost activities. Let Π_{i0} be aggregate profits net of all fixed costs and D_0 aggregate domestic final-good sales. Set

$$E_{h0} = L_h + \Pi_{d0} + \lambda_f \Pi_{f0}, \qquad F_0 = E_{h0} - D_0,$$
$$P_{h0} = \left(\frac{c_\sigma E_{h0}}{B_{h0}} \right)^{1/(1-\sigma)}. \tag{E9}$$

Here F_0 is baseline final-good import expenditure. The baseline requires $F_0 > 0$, positive expenditure and operation costs, and $\kappa > \sigma - 1$ for finite revenue moments. Record aggregate baseline exports X_0 using the same conditional weights and economic masses.

2. Evaluate firm choices and aggregate outcomes. For a given s, the joint solver varies

$$z = (\log \underline{\varphi}_d, \log \underline{\varphi}_f, \log B_h, \log \eta)'. \tag{E10}$$

At each candidate z, update conditional productivities and active masses using (E6). Hold the foreign potentials fixed at their values for the given s, and update the exchange-rate-adjusted coefficients using (E5). Each integration node selects the most profitable of the four import/export choices following equation (11). Operating costs are common across these choices and hence do not affect their ranking, but they are subtracted in net profits.

Let X denote aggregate exports in home currency, I imported-input expenditure, D domestic final-good sales, and Π_i net profits by ownership. With $\lambda_d = 1$, household expenditure follows equation (15); expenditure and repatriated profits are

$$E_h = L_h + \Pi_d + \lambda_f \Pi_f, \qquad R = (1 - \lambda_f)\Pi_f. \tag{E11}$$

Final-good import expenditure is calculated directly from CES demand:

$$F = F_0 \frac{B_h}{B_{h0}} \eta^{1-\sigma}. \tag{E12}$$

This is the ratio form of $F = (B_h/c_\sigma)(\tau^M \epsilon)^{1-\sigma}$ under the normalized foreign FOB price index. It uses the same normalization as (E3).

3. Solve the four equilibrium residuals jointly. For each s, solve

$$\mathcal{R}(z;s) = \begin{bmatrix} (b_d - f_d^O)/f_d^O \\ (b_f - f_f^O)/f_f^O \\ (X - I - F - R)/E_{h0} \\ (E_h - D - F)/E_{h0} \end{bmatrix} = 0. \tag{E13}$$

The first two entries impose the stated boundary rule. The third imposes the trade balance condition in equation (14), and the fourth imposes goods-market clearing in equation (16), using the currency and CES normalizations stated above. Household expenditure already satisfies equation (15). Recover the price index directly from equation (6) as

$$P_h = \left(\frac{c_\sigma E_h}{B_h}\right)^{1/(1-\sigma)}. \tag{E14}$$

Thus expenditure and the price index do not require additional adjustment loops.

I use MATLAB's `fsolve` in log coordinates, with central finite differences and relative finite-difference step 10^{-3}. The configured step and function tolerances are 10^{-9} and 10^{-10}; work is limited to 150 iterations and 1,500 function evaluations per equilibrium. The independently evaluated acceptance criterion is

$$\|\mathcal{R}(z;s)\|_\infty \leq 10^{-5}. \tag{E15}$$

Accepted states must have finite residuals, positive expenditure and a finite real price index, and must respect the Pareto support and entrant-mass bounds. The solver exit flag is recorded, but a stalled-step flag is not automatically treated as failure if the directly evaluated equations satisfy the acceptance checks. Discrete trade choices can generate small jumps in aggregate outcomes.

Two additional checks are imposed. Let L^D be total labor demand computed from firm outcomes, and let

$$H = \sum_i M_i \sum_j \omega_{ij} p_{ij}^{1-\sigma} + \frac{c_\sigma F_0}{B_{h0}} \eta^{1-\sigma}. \tag{E16}$$

I require

$$\left|\frac{L^D - L_h}{L_h}\right| \leq 5 \times 10^{-5}, \qquad \left|\frac{P_h^{1-\sigma} - H}{H}\right| \leq 5 \times 10^{-5}. \tag{E17}$$

Before calibrating a shock, the no-shock solution is checked against the baseline values. These tests certify the numerical equations on the chosen grid; they do not establish integration

accuracy or uniqueness of equilibrium.

4. Calibrate the foreign-demand shock. For every accepted equilibrium, calculate

$$g_X(s) = \frac{X(s)}{X_0} - 1, \qquad g_X(s) = g_X^{\text{data}}. \tag{E18}$$

The current target is $g_X^{\text{data}} = 0.1178881$, interpreted as nominal aggregate export growth. Its estimation strategy follows Appendix E.1.

The algorithm first evaluates $s = 1$ and probes $s = 1.01$ for a measurable local response. For the positive-demand experiment it constructs an upper endpoint

$$\log s_k = 1.5^{k-1} \log(1.02), \qquad k = 1, \ldots, 16, \tag{E19}$$

until $[1, s_k]$ brackets the target. It then uses MATLAB's `fzero` on $\log s$, with a coordinate tolerance of 10^{-9} and limits of 60 iterations and 100 function evaluations. An objective value within 10^{-5} of the target is treated as zero, and the actual final export-growth error is checked against the same threshold. This is an absolute tolerance of 0.001 percentage point. The algorithm raises an error if an admissible bracket or accepted solution is not found; it does not replace a positive-demand experiment with a contraction.

Solutions are cached by the log shock. Each new equilibrium starts from the previously solved equilibrium closest in log-shock distance, or from the baseline if none is available. All firm outcomes, masses and residuals used for reporting correspond to the final accepted solution; there are no subsequent parameter updates.

5. Compute the reported outcomes. After calibration, compute welfare as E_h/P_h, real wages as $1/P_h$, and all aggregate outcomes using economic masses and conditional weights. Report the foreign market potential $s\Theta_{a0}^X$ separately from the export revenue coefficient $C_X = \epsilon^\sigma s\Theta_{a0}^X = C_{X0} s\eta^\sigma$. Their proportional changes are $s - 1$ and $s\eta^\sigma - 1$, respectively; an appreciation can lower the latter despite an increase in foreign market potential. Equal-productivity and equal-fixed-cost decompositions are evaluated only after convergence and hold the corresponding equilibrium environment and masses fixed. They are conditional firm-choice decompositions rather than separately solved general-equilibrium experiments. Any comparisons across weighted sales ranks are identified as distributional comparisons and are not interpreted as changes for persistent firm identities.